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Audio Deepfake Detection Using Deep Learning

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ABSTRACT

This study introduces an enhanced Siamese convolutional neural network (Siamese CNN) architecture with a novel StacLoss function and self-attention modules for efficient identification of audio deepfakes. Our module directly compares unprocessed original audio with modified audio by initially applying convolutional operations and dual branches to extract complex characteristics from raw audio signals. These operations are followed by residual connections, which enhance the network's performance. The self-attention modules are trained in a layered way alongside these fundamental layers to detect multi-headed attention within audio frames. The StacLoss output represents a customized version of the contrastive loss function. It aids the network in distinguishing between original and modified audios by minimizing the loss between pairs of original audio that have the same identity while maximizing the distance between manipulated audio samples and enhances the process of extracting features compared to standard techniques. The efficacy of the method has been verified by examining a range of audio modifications, and its resilience has been thoroughly assessed on the ASVspoof2019 dataset by comprehensive testing across all possible audio manipulation situations. The proposed Siamese convolutional neural network (CNN) outperformed both machine and deep learning models, achieving impressive metrics. It achieved a remarkable accuracy of 98%, precision of 97%, recall of 96%, *F*1 score of 96.5%, ROC-AUC of 99%, and an equal error rate (EER) of 2.95%.

1 | Introduction

Audio deepfake technology has emerged as a significant threat in the modern digital landscape, posing risks to security, privacy, and trust in media communications. As this technology advances, real-world incidents highlight its disruptive potential across various domains, making the urgency of effective detection methods increasingly clear.

One prominent example of the impact of audio deepfakes occurred during the Russia – Ukraine conflict, where a fabricated audio clip of Ukrainian President Volodymyr Zelenskyy calling for military surrender went viral on social media platforms [1]. This incident showcased how deepfakes can be weaponized to undermine public trust, spread misinformation, and influence geopolitical narratives. Such tactics amplify the psychological and social repercussions of warfare, raising concerns about the manipulation of public opinion through synthetic media.

In the financial sector, the consequences of audio deepfakes have been equally alarming. For instance, attackers used deepfake technology to mimic the voice of a company CEO, successfully convincing employees to transfer approximately €220,000 to fraudulent accounts [2]. This incident underlines the vulnerability of voice-driven systems, particularly in contexts where voice authentication or verbal confirmation plays a crucial role. These cases also highlight the inadequacy of traditional security measures against emerging threats.

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Moreover, audio deepfakes are increasingly targeting the credibility of political figures, celebrities, and organizations. Manipulated audio recordings have been used to fabricate controversial statements or create false evidence in legal and reputational disputes [3]. For instance, in a widely circulated deepfake, former President Barack Obama was depicted making offensive remarks in a fabricated public service announcement [4, 5]. Such manipulations erode trust in digital communications and exacerbate challenges in verifying the authenticity of information.

The accessibility of deepfake tools has further exacerbated these challenges. Modern audio deepfake software requires minimal technical expertise, enabling widespread misuse. Platforms offering voice cloning services allow users to generate convincing synthetic voices with just a few seconds of input data, democratizing the creation of deceptive media [1, 6]. This accessibility has led to an increase in phishing scams, fake interviews, and fraudulent customer service calls.

The ethical and societal implications of these incidents are profound. Audio deepfakes not only compromise individual and organizational security but also pose broader risks to societal cohesion and trust in democratic processes [7]. Their potential for misuse necessitates a robust, interdisciplinary response involving researchers, policymakers, and technology developers.

Recent advancements in deep learning have enabled the development of robust models for diverse classification tasks, such as disease diagnosis and toxic tweet detection. For example, CNN-based models optimized with gray wolf optimizer (GWO) have shown high accuracy in classifying complex visual patterns, such as monkeypox skin lesions [8]. Similarly, language models like AraBERT demonstrate exceptional accuracy in classifying toxic Arabic tweets, showcasing the ability of transformer-based architectures in handling diverse data modalities [9]. These methodologies inform our approach, particularly in leveraging domain-specific feature extraction and model optimization.

Furthermore, innovative frameworks integrating advanced machine learning with cloud and blockchain technologies have been proposed for secure and efficient classification tasks, such as cancer diagnostics [10]. While such techniques have focused on medical imaging and text analysis, the field of audio processing remains underexplored. The application of self-attention modules within Siamese CNN architectures for audio deepfake detection seeks to address this gap.

Traditional detection approaches encompass both machine learning and deep learning techniques [11]. However, many existing methods still struggle with key challenges, such as generalizing across different audio domains and handling a variety of synthetic manipulations. Common deep learning approaches include long short-term memory (LSTM) networks and convolutional neural networks (CNNs) [12], which have proven to be effective in some cases. Nevertheless, these methods often present significant disadvantages, including:

- **Extensive Pre-processing:** Traditional detection approaches rely heavily on intensive feature extraction and pre-processing steps. These steps, while beneficial for accuracy, can delay detection, which is critical in real-time applications.
- **Feature Engineering:** Classical approaches depend on manual feature engineering, necessitating deep domain expertise. The requirement for manual feature extraction limits scalability and adaptability to new types of attacks.
- **Generalization Challenges:** One of the most critical issues in machine learning models is their ability to generalize to unfamiliar data. In audio deepfake detection, models that fail to generalize are prone to attacks. Given that humans are also susceptible to deepfakes, the challenge of detection becomes even more complicated.

Traditional detection models, while effective in some scenarios, face issues with scalability, generalization to unfamiliar data, and computational efficiency in real-time applications. Moreover, existing contrastive loss functions may not fully capture the subtle inter-class variations critical for distinguishing synthetic from authentic audio.

With the rapid advancement of deepfake technologies, researchers are increasingly focusing on more robust solutions that are both scalable and efficient in real-world scenarios. A key aspect of this is developing models that can handle a wider variety of deepfake generation methods while reducing computational complexity.

Previous research has demonstrated the effectiveness of deep learning techniques for audio deepfake detection. However, even these state-of-the-art models often require extensive computation and struggle with generalization to previously unseen data. This limitation is particularly evident in systems that are highly specialized or domain-specific. The rapid evolution of audio deepfake algorithms further exacerbates the problem, as new techniques emerge that can bypass existing detection methods. As such, continuous research is necessary to address these shortcomings.

This study aims to build an audio deepfake detection model that not only reduces the need for preprocessing but also achieves higher accuracy while maintaining scalability. The model leverages stronger features extracted from input speech and reduces the model size to enhance the detection mechanism. By focusing on minimal preprocessing, we aim to create a model that can efficiently process raw audio signals and detect manipulated audio in real time. Additionally, the model's architecture is designed to capture subtle variations in audio data that may indicate tampering, further improving detection accuracy.

1. **Siamese CNN Architecture with Self-Attention and Residual Connections:** We implemented a Siamese CNN architecture enhanced with self-attention modules and residual connections. This architecture significantly improves feature extraction, allowing the model to capture complex audio manipulations more effectively.
2. **StacLoss—A Customized Contrastive Loss Function:** To further enhance the model's precision, we propose StacLoss, a tailored contrastive loss function that helps the classifier distinguish between authentic and fake audio with greater accuracy.

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3. **Comprehensive Analysis Using ASVspoof2019 Dataset:** The ASVspoof2019 dataset is a widely accepted benchmark for evaluating audio deepfake detection methods. We conducted extensive testing across a range of audio manipulation situations to validate the effectiveness of our model. The results showed superior performance, both in terms of accuracy and robustness, when compared to other state-of-the-art models.

The complexity of audio signals makes them particularly difficult to analyze, especially in cases of deepfakes where subtle manipulations can make detection extremely challenging. This complexity is compounded by the fact that audio deepfakes are generated using increasingly sophisticated techniques, including autoregressive models and variation auto encoders, which blur the line between real and synthetic audio. Traditional methods that rely on hand-crafted features such as MFCCs (Mel-Frequency Cepstral Coefficients) and spectral analysis often fail to capture these subtleties.

To address these issues, the deep learning community has shifted toward models that can learn feature representations automatically from raw data, thus bypassing the need for extensive feature engineering. Our study follows this trend by implementing a self-attention mechanism that helps the network focus on important regions of the audio signal while ignoring less relevant parts. This not only improves the accuracy of detection but also reduces the computational burden, making the model more suitable for real-time applications.

In summary, the enhanced Siamese CNN architecture presented in this paper represents a significant advancement in the field of audio deepfake detection. By focusing on feature extraction and minimizing preprocessing, our model is better equipped to handle a wide range of deepfake scenarios, ensuring both robustness and efficiency in practical applications. The introduction of StacLoss further contributes to the model's success, helping it outperform existing methods in detecting audio deepfakes with greater precision. Through comprehensive testing on the ASVspoof2019 dataset, the proposed model demonstrates superior accuracy, making it a valuable tool for combating the growing threat of audio deepfakes.

The proposed model addresses the pressing need for a robust, efficient audio deepfake detection solution by leveraging minimal preprocessing and optimized feature extraction.

2 | Related Work

Audio Deepfake Creation Techniques can be classified into two ways. The first is TTS (text-to-speech) Synthesis: Speech synthesis uses deep learning to generate speech from text. Models like WaveNet [13] and Tacotron 2 [14] create realistic voices by training on extensive speech databases. Early TTS systems relied on audio concatenation methods, which lacked clarity compared to advanced deep learning techniques.

Recently, deep learning breakthroughs have widened the domain; for example, Gaussian neural audio encoders, autoencoders, and autoregressive models [15, 16]. Recently, interactions between humans and computers have witnessed significant advancements due to WaveNet's innovations, which generate audio from spectrograms [17]. Parallel WaveNet, an innovation, is an automated technique that improves sampling efficiency [18].

Deep Voice1 modularizes neural network templates for audio sources and text processors [19]. Google's Tacotron synthesizes audio from text and audio pairings, integrating sequence-to-sequence models and attention-based decoding but requires high-performance systems and repeated processing steps.

The second category is **Voice Conversion**: Voice conversion technologies modify speech to resemble another speaker while preserving grammar and adjusting the tone and pitch of speech. Various methods, such as Deep Voice [19], StarGAN-VC [20], and VC [21], are used in applications like expressive synthesis, personalized voice assistants, and dubbing.

Recent developments in anti-spoofing countermeasures for speaker recognition systems focus on generating fake data for VC systems [22, 23]. Most solutions are based on manipulating timbre and prosody, including pitch, amplitude, stress, and duration.

Voice Conversion techniques address data limitations, showing promising results for cross-lingual voice conversion [24].

Voice conversion strategies often rely on spectral mapping using paired data, where samples from both the target and source have identical content. Methods include Gaussian mixture models (GMM) [25] and parallel spectral modeling [26]. These approaches modify spectral features to capture temporal correlations, with RNN-based speaker-dependent methods also being proposed.

In addition to creation techniques, various methods exist for detecting audio deepfakes. These techniques can be classified based on several factors, including classical machine learning and deep learning approaches.

Classification models in diverse domains often face challenges related to data variability and computational efficiency. For instance, recent work on poultry audio signal classification introduced a novel Burn Layer for enhanced robustness against input variability [27]. Such approaches align closely with the goals of audio deepfake detection, where model resilience to manipulated signals is critical. Similarly, advancements in cancer prediction have utilized dimensionality reduction and optimization techniques to improve accuracy while reducing computational overhead [28, 29].

The integration of attention mechanisms, such as those employed in cardiovascular disease prediction using transformer-based models [30] provides a blueprint for enhancing feature extraction in audio deepfake detection. Moreover, research on leveraging feature selection for hepatocellular carcinoma highlights the role of automated optimization in improving model generalizability [28]. These insights guided the incorporation of self-attention modules and loss function enhancements in the proposed system.

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2.1 | Classical Machine Learning Methods

The key to these methods is feature extraction from the audio data in a handcrafted manner; for example, the Constant-Qefficients (CQFrequency CepstrMFCCs), and then feature classification by Logistic Regression model or support vector machines (SVMs) [31, 32]. As demonstrated by [33] the half-truth audio detection (HAD) dataset contains half-fake audio in which some words of an utterance are "watermarked" with synthetic speech production technology. The dataset also comes with fully fake and fully real audio. The authors ran benchmark experiments using the well-established classical machine learning techniques (GMM) and (LCNN) to detect half-truth and fully fake audio. Half-truth audio detection is much more challenging than fully fake audio detection because only portions of the audio are manipulated.

2.2 | Deep Learning Techniques

Deep learning models, especially CNNs and LSTMs [34, 35], have shown significant promise in audio deepfake detection. These models can learn intricate representations directly from raw audio data, potentially reducing the need for extensive preprocessing. Thai et al. [36] explores solutions for both audio deepfake detection and speech recognition for low-resource languages using deep learning. The audio deepfake detection part of the research utilizes convolution-recurrent neural networks (CRNNs) and convolutional-based methods to identify and counteract spoofing attacks, specifically those involving TTS and VC techniques. The methods are assessed using the ASVspoof 2019 Logical Access dataset, demonstrating significant improvements in identifying spoofed audio produced by various sophisticated algorithms. The study highlights the effectiveness of deep learning models in distinguishing between genuine and fake audio. This research contributes to the field of audio forensics and cybersecurity. Yang et al. [37] introduced a robust audio deepfake detection system that improves generalization through multi-view feature incorporation. They evaluated various audio features, including handcrafted (e.g., MFCC, CQT) and learning-based features (e.g., wav2vec 2.0, HuBERT). By employing feature selection and feature fusion mechanisms, their approach demonstrated superior generalization across datasets, achieving an equal error rate (EER) of 24.27% on the In-the-Wild dataset, making it a reliable solution for detecting deepfakes generated by diverse systems. In a broader context, Xie et al. [38] extended the detection scope beyond speech deepfakes to include general audio deepfakes through their FakeSound dataset. Their model focuses on detecting half-truth audio, where only parts of the audio are manipulated, making detection particularly challenging. The benchmark system outperformed human detection accuracy and surpassed state-of-the-art models, demonstrating the importance of considering non-speech deepfakes for a comprehensive detection strategy.

2.3 | End-to-End Models

In [39], Hua et al. demonstrated an end-to-end synthesized voice recognition model called Res-TSSDNet. It computes the deep features for detection in an efficient manner with a higher computational cost but good generalization to unseen samples. In [40], Zhang and Koishida introduce an all-encompassing system for speaker verification that is independent of text specifications and for short spoken phrases. In this work, they use a modified Inception Net with residual blocks to optimize the triplet-loss function. The input spectrogram is a fixed-size representation of an arbitrary length utterance. The proposed system consistently shows astonishing results compared with conventional i-vector-based models. It is particularly better in short duration conditions that range from 2 to 4 s.

2.4 | Hybrid Models

Jiang et al. [41] This model mimics PASE+ by being a self-supervised deep fake audio detector that extracts context features from audio streams using multilayer convolutional blocks. Therefore, it can be used to scale and operate efficiently. Al-Adwan et al. [42] This paper suggests a new approach of detecting deepfake media by using both CNN and recurrent neural networks (RNN), which use particle swarm optimization (PSO) for optimization. The CNN extracts spatial features from the video frames, and the RNN extracts sequential features. PSO is used to optimize the parameters of the model. This enhances the accuracy of the identified deepfake media. The suggested method shows perfect accuracy, sensitivity, and specificity on two datasets: Celeb-DF and Deepfake Detection Challenge (DFDC), as well as when compared with some latest and advanced algorithms. The model performs very well and has demonstrated good generalization among various datasets.

2.5 | Data Augmentation and Transfer Learning

A technique utilizing convolutional RNNs for audio deepfake classification was proposed by Chintha et al. [36], while Shan and Tsai [35] used alignment-based approaches with LSTM and transformer architectures. These methods can improve both the accuracy and robustness of models by combining transfer learning (or other fusion approaches) with data augmentation. Recently, a transfer learning approach for audio deepfake detection using a VIB with the wav2vec 2.0 pre-trained model was introduced by Eom et al. [43]. Their approach is aimed at improving generalization in the detection of spoofed speech by compressing the original signal based on what is important. Their model shows substantial improvements compared with the best-performing techniques in the ASVspoof 2019 database and also shows robustness in many scenarios with limited resources and using many datasets, which indicate their robustness in many data conditions. Zhu et al. [44] developed a novel approach known as SLIM (Style-Linguistics Mismatch), which enhances generalization by identifying mismatches between style (voice characteristics) and linguistic (verbal content) features in audio. This model consists of two stages: the first stage learns the style-linguistics dependency on real speech data using self-supervised learning, while the second stage uses supervised learning to classify real and fake audio. SLIM demonstrated superior performance on out-of-domain datasets and provides interpretability by identifying where style and linguistic features diverge. This method addresses a key challenge in audio deepfake detection, where models struggle with unseen attacks.

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2.6 | Generative Adversarial Networks

Generative adversarial networks (GANs) are used to generate realistic-sounding phony audio, which is in turn difficult to detect, and have been used to create sophisticated deepfakes based on GAN-based methods such as those of Wu et al. [45], Zhang et al. [46], and Li et al. [47].

2.7 | Models Based on Transformers

For detecting audio deepfakes, Transformers have also been employed. Vaswani et al. [48] introduced the Transformer architecture that has been adapted for suitability in detection tasks that involve sequential data like audio. Similarly, Huang and Pun [49] introduced a new framework that incorporates a self-attention mechanism, one of the primary components of Transformer architectures, for replay and Deep-Fake audio detection. Using this approach, the system makes use of hybrid features obtained by using convolutional neural networks (CNNs) and Mel-spectrograms. This enables the network to distinguish authentic sounds from falsified ones. Additionally, a ResNet classifier is employed to improve system performance in detecting spoofing attacks. Their approach displays state-of-the-art performance in detecting audio spoofing attacks in ASVspoof 2021 [50].

2.8 | Contrastive Learning

To learn representations that are robust against manipulation attacks, contrastive learning has proven to be particularly effective in learning these representations. Based on techniques proposed in [51], the authors in [52] propose CLAD (Contrastive Learning-based Audio deepfake Detector) to learn robust audio representations for manipulations in audio-deepfake detection models. CLAD formulates audio deepfake detection as a contrastive learning problem. Utilizing unpaired data in training, it focuses on learning from a positive-pair similarity constraint to distinguish similar and dissimilar pairs. Moreover, the authors introduce a technique of length reduction that achieves an additional boost in precision. This is done by pushing genuine audio features that are spatially close in the feature space together. This system significantly improves the system's robustness by achieving a FAR below 1.63% over all types of manipulations with accuracies exceeding the best performance in the current literature.

2.9 | Ensemble Models

Researchers such as Chettri et al. [53] and Deng et al. [54] have also worked on ensemble models where multiple base models are collected in a single system to improve the detection performance of each model. In this way, these systems leverage the benefits of individual models to enhance the overall accuracy and robustness. Pham et al. [55] have proposed an audio deepfake detection method based on deep learning that uses an ensemble of models consisting of different spectrograms. Here, the recorded audio is first converted into spectrograms using three ways: Short-time Fourier Transform (STFT), which enables the study of the frequency content of a signal over a short time interval; Constant-Q Transform (CQT), which does not produce distortion while compressing the spectrum; and Transformation of wavelets (WT) using different auditory filters. The data is then fed into three different deep learning models: First, the difference between an original audio file and its spectrogram is directly fed into a CNN model to train the system; then, the transfer learning approach which utilizes state-of-the-art computer vision models; After that, the audio embeddings using pre-trained models such as Whisper and SpeechBrain. Finally, the best-performing model from each approach is merged to form an ensemble, which achieves an (EER) of 0.03 when tested on the dataset of ASV Spoof 2019, thereby demonstrating the high effectiveness of audio deepfakes.

2.10 | Siamese CNN Applications

Siamese CNNs have been widely used in similarity or dissimilarity comparison tasks. Zhenchun Lei et al. [56] Proposed a hybrid model that combines a 1D-CNN for feature extraction with a Siamese CNN for classification, resulting in optimal detection. While Nekadi [57] proposes an intricate solution for an in-depth analysis of deepfake content. This approach employs a Siamese CNN with several modes of analysis to detect deepfakes. The method focuses on the correlation between the visual and auditory senses in the videos to detect deepfakes. The descriptions of frames are embedded using Siamese networks. The model computes the similarities of the embedded data of genuine and fake videos. The detected facial features are extracted from the video using OpenCV, while the audio features were extracted using the LSTNet architecture. The model achieved improved performance in the detection of deepfake videos using different data sets such as FaceForensic++, DFDC, CelebDF, with higher accuracy. The effectiveness of using Siamese networks for deepfake detection was achieved using contrastive and triplet loss functions to minimize the distance between the embeddings of genuine videos and maximize the distance between the embeddings of fake videos.

2.11 | Challenges and Limitations in Traditional and Deep Learning-Based Approaches

Despite notable advancements in audio deepfake detection, both traditional machine learning (ML) methods and deep learning (DL) models encounter fundamental challenges that hinder their effectiveness in real-world applications. While ML models rely on handcrafted features for detection, making them inherently domain-specific, DL-based techniques, though more flexible, remain computationally expensive and vulnerable to adversarial manipulations.

2.12 | Limitations of Traditional Machine Learning Approaches

Traditional ML-based methods, such as SVMs, GMMs, and Decision Trees (DTs), are commonly used for deepfake detection. These techniques require feature engineering, where audio characteristics such as Mel-Frequency Cepstral Coefficients

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(MFCCs), Linear Predictive Coding (LPC), and spectral properties are manually extracted and used for classification [58].

However, these approaches suffer from poor generalization, primarily due to their reliance on static feature sets. Deepfake generation techniques continuously evolve, incorporating new synthesis models, adaptive voice modulation, and noise-resilient artifacts, which traditional ML models fail to capture. This leads to:

- **High False Positive and False Negative Rates**: A model trained on specific attack types struggles with unseen manipulations, resulting in unreliable classification.
- **Sensitivity to Variability in Audio Quality**: Environmental factors, such as background noise, reverberation, and microphone distortions, degrade feature consistency.
- **Scalability Limitations**: The necessity of manual feature selection and tuning makes these methods inefficient for large-scale or real-time applications.

2.13 | Deep Learning Models: Strengths and Vulnerabilities

Deep learning models, particularly CNNs, RNNs, and hybrid architectures (e.g., Convolutional-Recurrent Neural Networks—CRNNs), have significantly improved deepfake detection accuracy by learning data-driven representations directly from audio waveforms or spectrograms [59]. These models eliminate the need for manual feature extraction, offering higher adaptability to diverse datasets.

Despite their success, deep learning-based methods remain vulnerable to evolving deepfake techniques due to several key issues:

Adversarial Susceptibility—Many deep learning classifiers can be fooled by imperceptible perturbations, making them ineffective against adversarially crafted attacks. Studies have shown that even minor waveform alterations can cause state-of-the-art models to misclassify synthetic audio as real [60].

Dataset Bias and Overfitting—Most deep learning models are trained on benchmark datasets such as ASVspoof2019 and FakeAVCeleb, which may not fully represent the diversity of real-world deepfakes [61]. When exposed to unseen data, these models often fail to generalize, leading to reduced robustness against previously unknown spoofing techniques.

Challenges in Detecting Hybrid Deepfakes—Some manipulations blend real and synthetic speech, making them significantly harder to detect. Unlike fully synthetic deepfakes, hybrid deepfakes retain natural voice characteristics, requiring models to distinguish subtle inconsistencies that are often overlooked by conventional DL-based classifiers.

Computational Constraints—Deep learning models, particularly transformer-based architectures and large CNNs, demand high computational power. This presents a challenge for real-time and resource-limited environments, such as mobile or embedded security applications.

2.14 | Ongoing Challenges and Future Research Needs

As deepfake generation methods become more sophisticated, detection models must continually evolve. Future research should explore:

Multi-Modal Fusion Strategies—Integrating acoustic, linguistic, and biometric signals for enhanced deepfake detection.

Robust Generalization Techniques—Developing self-supervised and contrastive learning frameworks to reduce dataset bias.

Adversarial Training—Training detection models against adversarial attacks to increase resilience.

Lightweight Architectures for Real-Time Use—Optimizing deepfake detection for low-power devices and real-time systems.

2.15 | Addressing These Limitations With Siamese CNN and StacLoss

While previous studies have employed Siamese CNNs [62] for similarity-based deepfake detection, our work introduces key innovations to address the aforementioned challenges:

Self-Attention Modules—Enhance the model's ability to detect fine-grained temporal variations in manipulated audio.

StacLoss (Stacked Contrastive Loss)—Improves inter-class separation and minimizes intra-class variability, making detection more reliable.

Efficiency and Scalability—Reduces reliance on preprocessing and handcrafted features, optimizing performance for large-scale applications.

3 | Methodology

3.1 | Siamese CNN Architecture With Self-Attention and Residual Connections

The proposed **Siamese CNN** consists of two **identical** branches, each processing an input audio sample separately. The architecture follows a structured pipeline:

3.1.1 | Audio Feature Extraction

- The input audio is converted into **Mel-Frequency Cepstral Coefficients (MFCCs)**, a widely used technique in speech and deepfake detection.
- The extracted MFCCs are **normalized and reshaped** before being fed into the Siamese CNN.

3.1.2 | Siamese CNN Structure

Each branch of the Siamese CNN follows the same architecture to ensure consistency in feature extraction. The configuration is as described in Table 1:

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TABLE 1 | Key parameters of the proposed Siamese CNN layers.

Layer type	Configuration	Activation function
Input layer	2D MFCC Features (40 coefficients)	N/A
Conv layer 1	32 filters, kernel size = (3 × 3), stride = 1	ReLU
Max pooling 1	Pool size = (2 × 2)	N/A
Conv layer 2	64 filters, kernel size = (3 × 3), stride = 1	ReLU
Max pooling 2	Pool size = (2 × 2)	N/A
Conv layer 3	128 filters, kernel size = (3 × 3), stride = 1	ReLU
Residual connection	Skip connection between Conv Layer 2 & 3	N/A
Self-attention module	Multi-Head Attention Mechanism (8 heads)	Softmax
Dropout layer	Dropout rate = 0.5	N/A
Fully connected layer	128 neurons	ReLU

3.1.3 | Feature Similarity Computation and Classification

- The embeddings generated by both CNN branches are compared using **Euclidean distance**.
- The **StacLoss function** optimizes similarity scores by:
  - **Minimizing** distance for **genuine** audio pairs.
  - **Maximizing** distance for **fake** audio pairs.
- A **threshold (T)** is used to classify the input as **real or fake**.

3.1.4 | Design Considerations

- **Self-Attention** enhances feature learning by allowing the model to focus on **important spectral and temporal regions**.
- **Residual connections** improve **gradient flow**, leading to more stable training.
- **Dropout regularization** prevents overfitting, ensuring **robust generalization**.

A detailed diagram illustrating the architecture is provided in Figures 1 and 2.

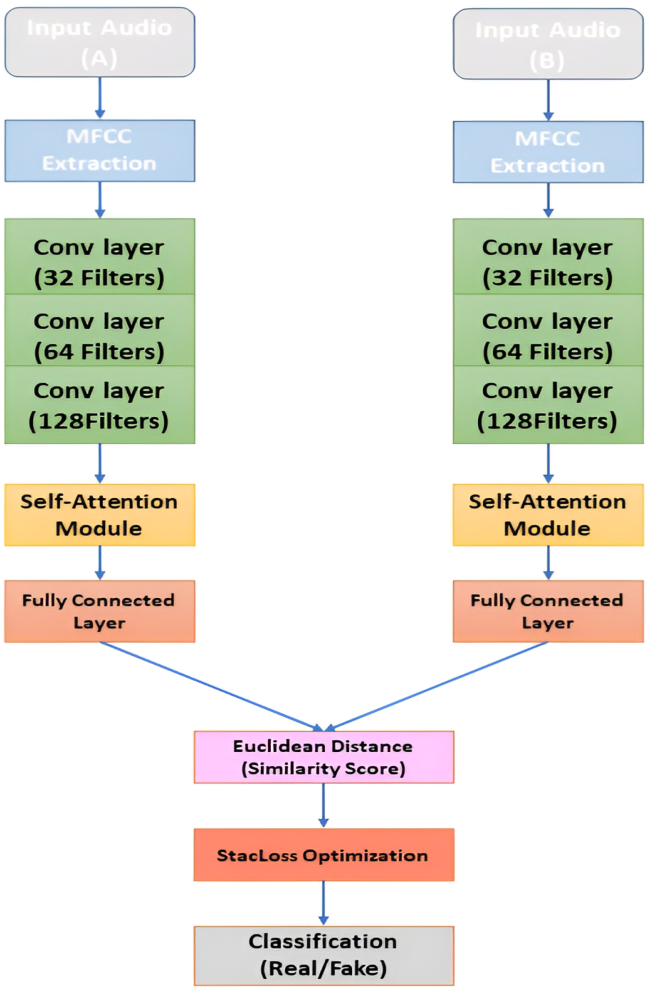

FIGURE 1 | The architecture of the proposed Siamese CNN for audio deep fake detection.

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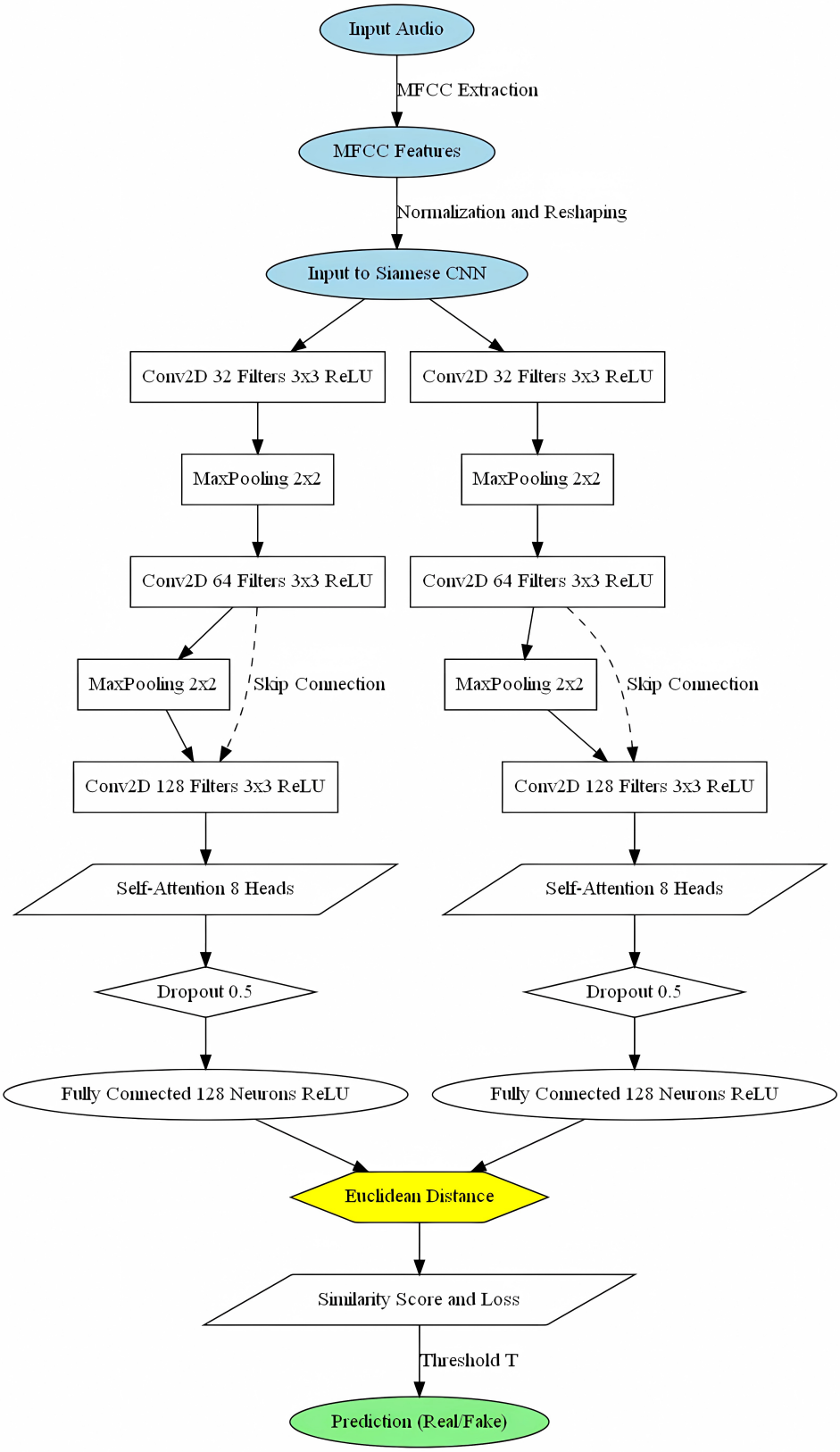

FIGURE 2 | Detailed Siamese CNN architecture for audio deepfake detection.

Figure 1 provides a high-level overview of the Siamese CNN framework for audio deepfake detection, illustrating the main processing stages from input feature extraction to final classification.

Figure 2 illustrates the detailed architecture of our Siamese CNN, including the specific number of convolutional layers, filter sizes, activation functions, and additional components such as self-attention and dropout layers.

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TABLE 2 | Hyperparameter settings and experimental configurations.

Parameter	Value	Description
Optimizer	Adam	Adaptive learning rate optimization algorithm
Learning rate	$1e^{-3}$	Final learning rate chosen after grid search
Batch size	64	Number of samples processed per training step
Epochs	100	Maximum number of training iterations over the dataset
Loss function	StacLoss	Customized contrastive loss function for similarity optimization
Filter sizes (Conv)	[32, 64, 128]	Number of filters in convolutional layers
Kernel size	3×3	Size of the convolutional kernels
Pooling	Max pooling (2×2)	Pooling strategy for dimensionality reduction
Dropout rate	0.3	Fraction of units dropped to prevent overfitting
Fully connected neurons	128	Number of neurons in the fully connected layers
Dataset used	ASVspoof2019	Benchmark dataset for evaluation
Input features	MFCC (13 coefficients/frame)	Feature extraction technique for input audio
Attention module	Self-attention	Enhances feature focus in Siamese branches

3.2 | Triplet Selection and Training Parameters for Siamese Network

In our proposed audio deepfake detection method utilizing a Siamese CNN, we employ a triplet-based training approach to learn discriminative embeddings. Each triplet comprises:

- **Anchor (A)**: An audio sample from a specific speaker.
- **Positive (P)**: Another genuine audio sample from the same speaker as the Anchor.
- **Negative (N)**: A spoofed version of the Anchor's speaker or an audio sample from a different speaker.

Triplet Mining Strategy: We adopt the **Batch Hard Triplet Mining** strategy, which selects the hardest positive and negative samples within each batch for a given anchor. This ensures:

- **Hard Positive**: The positive sample with the maximum distance to the anchor.
- **Hard Negative**: The negative sample with the minimum distance to the anchor.

This approach ensures that the model focuses on the most challenging examples, thereby enhancing its discriminative capability.

Margin Value: A margin value of 1.0 is utilized in the triplet loss function. This margin defines the minimum acceptable distance between positive and negative pairs, encouraging the network to separate genuine and fake audio embeddings by at least this margin.

Training Hyperparameters: The training process is configured with the following hyperparameters:

- **Learning Rate**: Set to 0.001 to facilitate stable convergence during training.
- **Batch Size**: Chosen as 64 to ensure a sufficient number of triplets per batch, promoting effective learning.
- **Embedding Dimension**: The network outputs embeddings of 128 dimensions, balancing computational efficiency and representational capacity.

By incorporating these strategies and parameters, the Siamese CNN is trained to effectively distinguish between genuine and fake audio samples, enhancing the robustness of the audio deepfake detection system (Table 2).

3.3 | Proposed Algorithm Description

To provide a clear understanding of the proposed system, the pseudocode below and the flowchart in Figure 3 outline the key steps involved in the audio deepfake detection process using Siamese CNN with self-attention and the StacLoss function. The algorithm operates as follows:

```
Input: Dataset (D) containing audio samples;
Margin (m); Training Epochs (E); Thresh-
old (T)
Output: Trained Model; Classification Result
(Real/Fake)

Training Phase:
1. Initialize Model:
  - Siamese CNN branches with identical
    architectures.
  - Loss Function: StacLoss with margin m.
2. For each epoch e in E:
  a. Sample Triplets:
    - Anchor (A), Positive (P), and
      Negative (N) from dataset D.
  b. Preprocess audio signals A, P, N:
    - Features_A ← Extract_MFCC(A)
    - Features_P ← Extract_MFCC(P)
    - Features_N ← Extract_MFCC(N)
```


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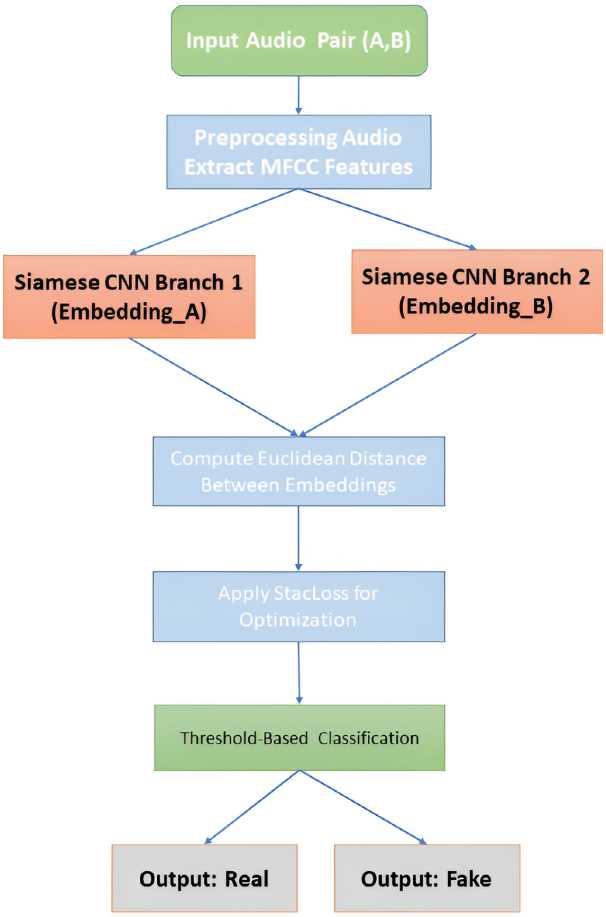

FIGURE 3 | Flowchart of the proposed audio deepfake detection method.

```
   c. Feature Extraction through Siamese CNN:
     - Embedding_A ← Process Features_A through
       Branch_1.
     - Embedding_P, Embedding_N ← Process
       Features_P, Features_N through Branch_2.
   d. Compute Distances:
     - d_pos ← Euclidean_Distance(Embedding_A,
       Embedding_P)
     - d_neg ← Euclidean_Distance(Embedding_A,
       Embedding_N)
   e. Optimize StacLoss:
     - Loss = max(0, d_pos - d_neg + m)
     - Update weights to minimize Loss.

Inference Phase:
1. Preprocess input audio pair (X, Y):
   - Features_X ← Extract_MFCC(X)
   - Features_Y ← Extract_MFCC(Y)
2. Feature Extraction:
   - Embedding_X ← Process Features_X through
     Branch_1.
   - Embedding_Y ← Process Features_Y through
     Branch_2.
3. Compute Similarity:
   - Similarity_Score ← Euclidean_Distance
     (Embedding_X, Embedding_Y)
4. Classify:
   - If Similarity_Score ≤ T:
       Classify as Real.
   - Else:
       Classify as Fake.
5. Output Result.
```

Explanation of Key Components:

1. **Input Preprocessing:** Two audio signals are transformed into Mel-Frequency Cepstral Coefficients (MFCC) to capture their time-frequency representation.
2. **Feature Extraction:** Each signal passes through identical Siamese CNN branches. Convolutional layers extract features, while self-attention modules enhance critical regions of the feature map.
3. **Similarity Measurement:** The Euclidean distance between embeddings quantifies similarity.
4. **Loss Function Optimization:** StacLoss minimizes intra-class distance for real pairs and maximizes inter-class distance for fake pairs.
5. **Classification:** A threshold is applied to determine the class (real/fake).

3.4 | Dataset Description

The **ASVspoof 2019** dataset was employed to evaluate the proposed Siamese CNN-based audio deepfake detection method. This dataset is a widely recognized benchmark for spoofing detection tasks, containing a balanced distribution of real and fake audio samples. It encompasses diverse spoofing techniques, including voice conversion, speech synthesis, and adversarial examples, ensuring comprehensive testing across multiple attack scenarios.

The dataset is structured into two primary partitions:

1. **Logical Access (LA)**
2. **Physical Access (PA)**

Each partition is further divided into three subsets:

- **Training Set**
- **Development Set**
- **Evaluation Set**

As can be seen from Table 3, the audio files are sampled at 16 kHz, and features such as MFCC and Log-Spectrogram are extracted for processing. The dataset includes English and multilingual audio samples, ensuring diversity in language representation.

3.5 | Dataset Preparation and Experimental Setup

To test the generalization capabilities of the proposed Siamese CNN model, experiments were conducted using the

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TABLE 3 | Dataset features and splits.

Dataset feature	Logical access (LA)	Physical access (PA)
Training set	25,380 samples (2580 bonafide, 22,800 spoofed)	54,000 samples (5400 bonafide, 48,600 spoofed)
Development set	24,844 samples (2548 bonafide, 22,296 spoofed)	29,700 samples (5400 bonafide, 24,300 spoofed)
Evaluation set	71,237 samples (7355 bonafide, 63,882 spoofed)	171,000 samples (18,000 bonafide, 153,000 spoofed)
Total samples	121,461	254,700
Sampling rate	16 kHz	16 kHz
Features used	MFCC, Log-Spectrogram	MFCC, Log-Spectrogram
Languages represented	English and Multilingual	English and Multilingual

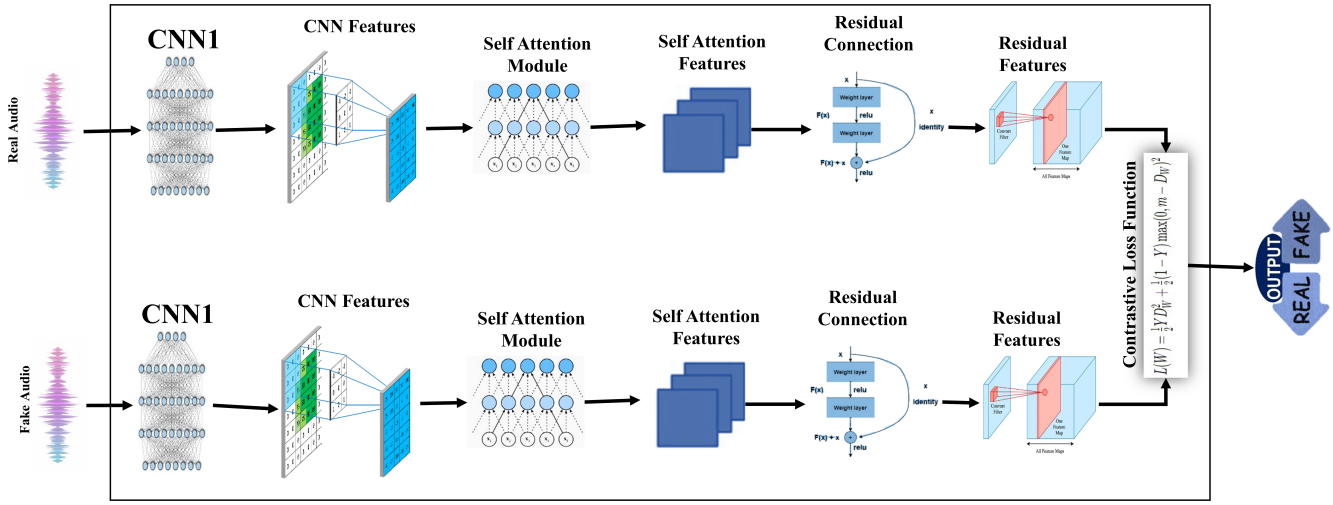

FIGURE 4 | Block diagram of the proposed Siamese CNN architecture.

FakeAVCeleb dataset. This dataset comprises diverse deepfake audio manipulations, making it a valuable benchmark for evaluating robustness. The dataset was split into training (70%), validation (10%), and test (20%) sets, ensuring non-overlapping distributions. All audio samples were preprocessed to extract MFCC features before being fed into the model. This experimental setup allowed for a fair and consistent evaluation across datasets.

3.6 | Justification of Design Choices

Design decisions, such as the use of self-attention modules and residual connections, alleviate some of the problems of the earlier methods, particularly in reducing the amount of preprocessing and improving feature extraction, leading to more robust and accurate detection.

The following diagrams illustrate the detailed block system structure for the proposed methodology:

As shown in Figure 4, the detailed arrangement of the Siamese CNN architecture, explaining how parallel processing of both real and fake audio samples occurs through the same CNN and self-attention modules in one branch, before being connected in a residual way and finally put through a contrastive loss function to check whether they are the same or not.

Figure 5 highlights the Siamese CNN architecture, which extracts features and reduces dimensions in a two-branch structure through pooling and dropout layers to reach fully connected representations by dense layers, and eventually calculates the contrastive loss as the final result.

This diagram in Figure 6 shows the overall workflow, from initial audio feature extraction using Mel-Frequency Cepstral Coefficients to the Siamese CNN processing and the contrastive loss function that determines the output.

3.6.1 | Mathematical Model

1. **Feature Extraction**: In the Siamese network, each subnetwork computes a feature vector representation, $A = f(x)$ for the input x:

2. **Euclidean Distance Calculation**: For a given pair of inputs, $(x1, x2)$, the Euclidean distance D_w between their feature vectors determined by:

$$D_w = \sqrt{\sum_{i=1}^{n} \left(\mathbf{f}(x_1)_i - \mathbf{f}(x_2)_i\right)^2} \tag{1}$$

where D_w: Euclidean distance between the feature vectors of x_1 and x_2. $\mathbf{f}(x_1)_i$, $(x_2)_i$: The ith components of the feature

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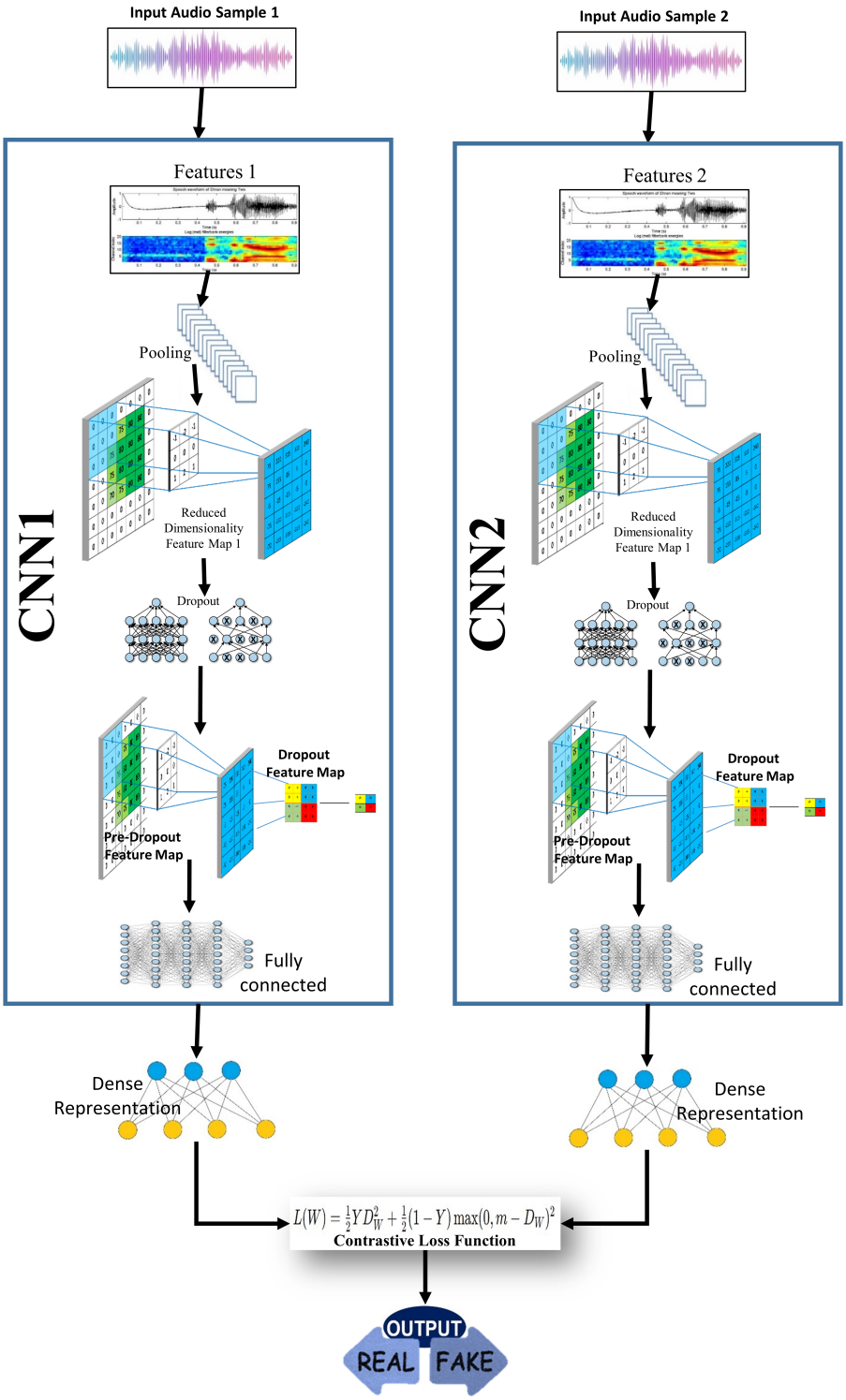

FIGURE 5 | Detailed view of the feature extraction process.

vectors for x_1 and x_2. n: Number of dimensions in the feature vectors.

3. **Convolutional Layer:** The convolutional operation, central to extracting local features from audio signals, produces an output feature map, represented as h_i

$$h_i = \sigma(W_i * x + b_i) \quad (2)$$

where h_i is the output feature map. W_i is the filter kernel. $*$ denotes the convolution operation. b_i is the bias term. σ is the activation function.

4. **Pooling Layer:** To manage computational complexity while preserving essential information, a pooling operation is applied, yielding a pooled feature map, p_i

$$p_i = \text{pool}(h_i) \quad (3)$$

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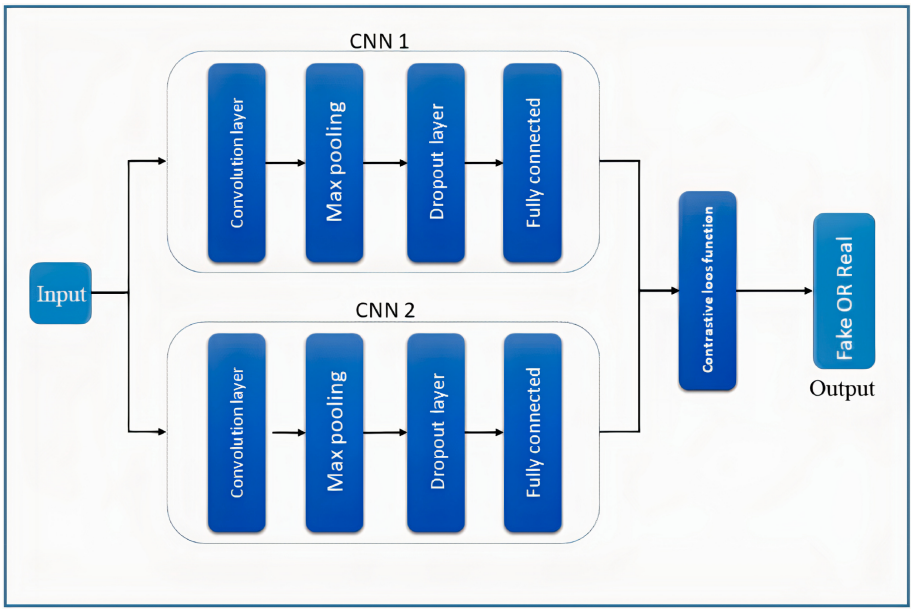

FIGURE 6 | Workflow of the proposed audio deepfake detection system.

where p_i: Pooled feature map after dimensionality reduction. pool: pooling function. h_i: input feature map from the convolution layer.

5. **Dropout Layer:** To prevent overfitting and improve generalization, a dropout operation is applied as follows.

$$d_i = \text{dropout}(p_i, \text{rate}) \tag{4}$$

where d_i output after applying dropout. dropout: dropout function, which randomly deactivates certain neurons. p_i: input pooled feature map. rate: dropout rate, specifying the fraction of neurons to drop.

6. **Self-Attention Modules**

Self-attention mechanisms allow the network to focus on different parts of the audio signal, capturing long-range dependencies and improving feature representations.

Given an input feature map **X**, the self-attention mechanism computes three matrices: queries **Q**, keys **K**, and values **V**. These are obtained through linear transformations:

$$\text{Q} = \text{W}_Q\text{X} \tag{5}$$

$$\text{K} = \text{W}_K \tag{6}$$

$$\text{V} = \text{W}_V\text{X} \tag{7}$$

where Q: query matrix derived from input X. K: key matrix derived from input X. V: value matrix derived from input X. W_Q, W_K, W_V: weight matrices for transforming X into Q, K, and V, respectively. X: input feature map.

The attention scores are computed using the scaled dot-product:

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right) \tag{8}$$

where **A**: attention score matrix. softmax: normalization function to convert scores into probabilities. **Q**: query matrix. $\mathbf{K}^T$: transpose of the key matrix. d_k: dimensionality of the key vectors, used for scaling.

The output of the self-attention layer is then calculated as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{AV} \tag{9}$$

where Attention($\mathbf{Q}, \mathbf{K}, \mathbf{V}$): output of the attention mechanism. which combines attention scores with values **V**.

To better illustrate how the self-attention mechanism identifies critical regions in audio signals, attention maps for real and fake audio samples are visualized in Figure 7. These maps highlight the temporal and spectral regions that the model focuses on while extracting features. For real audio signals, the attention mechanism emphasizes steady spectral regions corresponding to natural speech patterns. In contrast, for fake audio signals, the attention mechanism detects anomalies in temporal transitions and frequency distributions caused by synthesis artifacts. These visualizations demonstrate how self-attention enables the model to capture subtle differences, significantly improving feature extraction for classification.

7. **Residual Connections**

To facilitate the flow of information and gradients, aiding in the training of deeper architectures, residual connections are incorporated.

With Self-Attention: For self-attention, the residual connection output r_i is defined as:

$$r_i = p_i + \text{Attention}(Q, K, V) \tag{10}$$

where r_i: output of the residual connection with self-attention. p_i: pooled feature map. Attention(Q, K, V): result of the self-attention layer.

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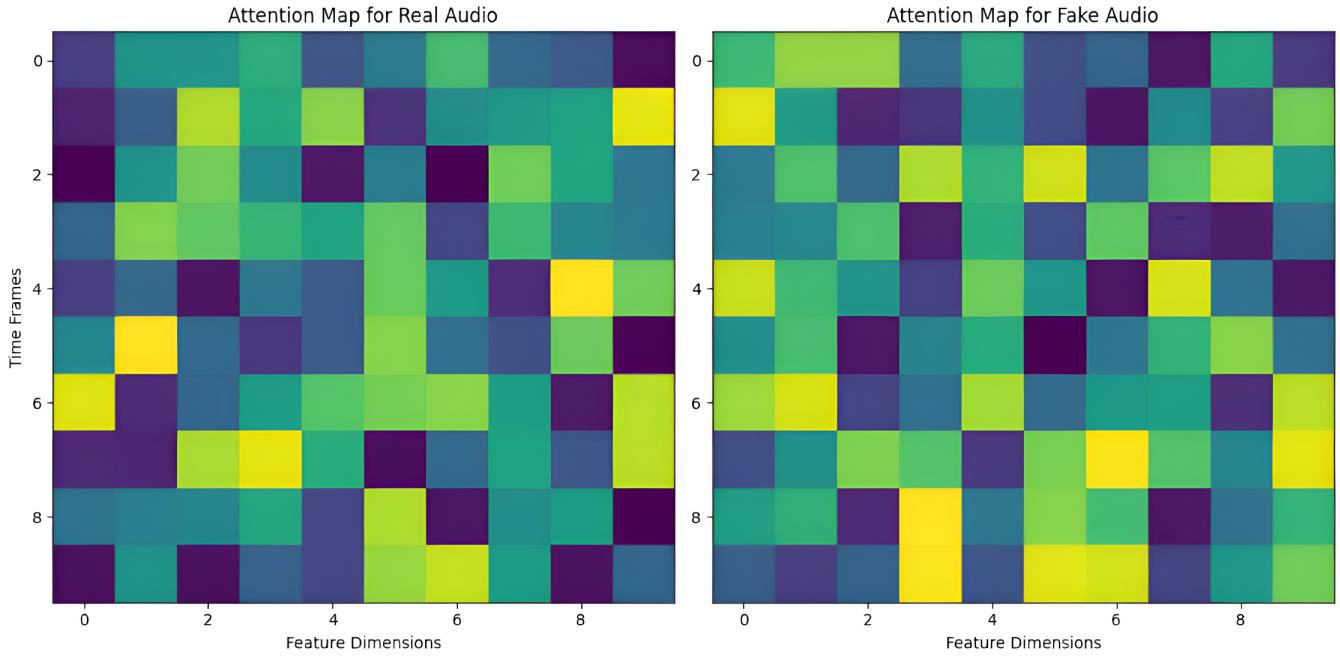

FIGURE 7 | Attention maps showing areas of focus for real and fake audio samples, visualizing the self-attention mechanism's impact on feature representation.

General Form: In a residual block, the output $\overline{Y}$ is calculated as follows:

$$\overline{Y} = \mathcal{F}\left(\mathbf{X}, \{W_i\}\right) + \mathbf{X} \tag{11}$$

where $\overline{Y}$ is the output of a residual block, $\mathbf{X}$ is the input, and, $\mathcal{F}\left(\mathbf{X}, \{W_i\}\right)$, represents a transformation function applied to $\mathbf{X}$. This structure helps in mitigating the vanishing gradient problem and supports the training of deeper networks.

8. **StacLoss: A Tailored Contrastive Loss Function**
   StacLoss is designed to measure the distance between feature vectors of input pairs, enabling the network to distinguish between real and fake audio samples. The StacLoss function, (W), is formulated as follows:

$$L(W) = \frac{1}{2} Y D_W^2 + \frac{1}{2}(1 - Y) max\left(0, m - D_W\right)^2 \tag{12}$$

where $L(W)$ is the loss function to be minimized. W represents the network's weights (parameters). D_W denotes the Euclidean distance between feature vectors of two audio samples. Y is a binary label indicating similarity (1) or dissimilarity (0). m is a margin parameter.

The objective of StacLoss is to minimize this loss function, encouraging similar pairs to have a smaller distance and dissimilar pairs to have a larger distance.

3.7 | Integration of Self-Attention With Siamese CNN

The self-attention module, mathematically described in the **Mathematical Model**, plays a pivotal role in enhancing the Siamese CNN's feature extraction capabilities. Within each CNN branch, the self-attention mechanism computes attention scores by dynamically weighing input features based on their relevance. This integration enables the network to focus on critical spectral and temporal regions of the audio signal, which are often indicative of deepfake manipulations.

Practically, the self-attention module is embedded after the final convolutional layer in each Siamese branch. It interacts with the convolutional feature maps, refining them by prioritizing key patterns while suppressing irrelevant information. This refinement allows the Siamese CNN to capture subtle anomalies and long-range dependencies in the audio, improving its robustness against complex manipulations. Quantitative metrics (e.g., attention-enhanced EER and t-DCF reductions) and qualitative visualizations (e.g., attention maps) provided in the Results section further validate the contribution of this mechanism.

3.8 | Advantages of the Proposed Method

Besides, the Enhanced Siamese CNN architecture has several advantages over traditional and other deep learning techniques:

The work described in this article pertains to the advanced detection of audio deepfakes, where an advanced Siamese CNN architecture has been used to address some of the limitations of the current audio deepfake detection techniques. The primary outcomes of this study are stated as follows:

3.8.1 | Development of Enhanced Siamese CNN Architecture

Our proposed architecture processes pairs of audio samples with two identical CNN subnetworks. This Siamese network architecture compares features extracted from authentic recordings to corresponding features extracted from manipulated audio samples.

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3.8.2 | Incorporation of Self-Attention Modules and Residual Connections

In order to enhance the understanding of characteristics and improve the ability to capture long-term dependencies in audio signals, the architecture incorporates self-attention modules and residual connections. This integration facilitates the network in acquiring a deeper understanding of complex interactions within the data.

3.8.3 | Introduction of StacLoss: A Tailored Contrastive Loss Function

A novel contrastive loss function, StacLoss, is used to train the Siamese CNN to identify genuine and artificial audio samples. StacLoss is designed to yield minimal distances for genuine audio pairs and large distances for manipulated ones.

3.8.4 | Reduction of Preprocessing Requirements

Unlike traditional methods, which require extensive preprocessing and feature engineering, this method reduces this work to a minimum by letting the model work directly with the raw audio signals, thereby simplifying the detection task.

3.8.5 | Comprehensive Evaluation on ASVspoof2019 Dataset

The effectiveness and robustness are further demonstrated on the ASVspoof2019 dataset and different types of artificial speech, which consist of both replay attacks as well as speech-face-transformation scenarios. The model achieves great improvements in detection accuracy and robustness.

4 | Experiments

4.1 | Datasets

The performance is evaluated with the aid of the ASVspoof 2019 dataset [22] containing a multitude of real/fake audio samples, which serves as a "golden standard" for evaluations.

4.2 | Data Curation

ASVspoof 2019 dataset contains audio deepfakes with different characteristics; this ensures a thorough examination. Data augmentation operations, such as time-stretching and pitch-shifting, are applied to increase the model's robustness.

4.3 | Training Process

The network was trained using a grid search to determine the optimal hyperparameters, including the learning rate, batch size (mini-batch size), and dropout rate. The training process stops based on an early stopping mechanism, either when the maximum number of epochs is reached or when the validation loss does not improve for 10 epochs.

The grid search was conducted over the following parameter ranges:

- Learning rate: $1e^{-4}$ To $1e^{-2}$
- Batch size: 32–128
- Dropout rate: 0.1–0.5

After tuning, the chosen hyper parameters were:

- Learning rate: $1e^{-3}$
- Batch size: 64
- Dropout rate: 0.3

To enhance training stability and ensure optimal convergence, the following **training strategies** were implemented:

1. **Optimization Algorithm:**
   - The **Adam optimizer** was employed for adaptive learning rate adjustments.
   - Configuration parameters: $\boldsymbol{\beta 1 = 0.9,\ \beta 2 = 0.999}$, ensuring effective momentum updates.
2. **Learning Rate Schedule:**
   - A **ReduceLROnPlateau** scheduler dynamically adjusts the learning rate.
   - If the validation loss does not improve for **5 consecutive epochs**, the learning rate is reduced by **a factor of 0.5**.
   - A minimum learning rate of $1e^{-6}$ prevents excessive reductions and stabilizes fine-tuning.
3. **Regularization Techniques:**
   - **L2 Regularization** (**Weight Decay** = $1e^{-4}$) is applied to convolutional layers to control model complexity and mitigate overfitting.
   - **Batch Normalization** ensures stable training by normalizing activations across layers.
   - **Dropout (0.3)** is applied to dense layers to improve generalization.

The training was conducted for a maximum of 100 epochs, with early stopping applied when there was no improvement in the validation loss for 10 consecutive epochs. The training process was performed on an NVIDIA Tesla V100 GPU (16 GB) using the Tensor Flow framework. Each training run took approximately 20 h, incurring an estimated computational cost of ˜$60 on cloud platforms like AWS or GCP. Figure 8 below illustrates the learning curves observed during the training process.

4.4 | Deployment Cost Analysis

Deploying deep learning models for audio deepfake detection involves both hardware and software considerations. Below is a summary of the estimated costs associated with training and deploying the proposed Siamese CNN model:

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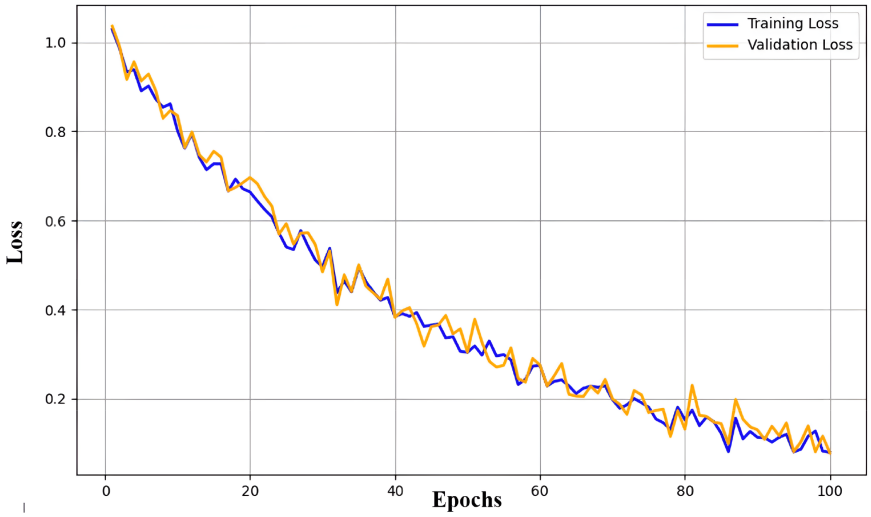

FIGURE 8 | Training and validation loss curves.

4.4.1 | Hardware Costs

- **Training Environment:** Training was conducted on an NVIDIA Tesla V100 GPU (16 GB) via cloud services such as AWS or GCP. Each training run required approximately 20 h, costing ~$3 per hour for GPU usage, leading to an estimated training cost of $60 per run.
- **Inference Environment:** Local deployment during inference can be achieved on more affordable hardware, such as the NVIDIA RTX 3080 ($700) or specialized edge devices like the NVIDIA Jetson Nano ($99) for lightweight applications.

4.4.2 | Software Costs

- **Frameworks and Libraries:** Development utilized open-source frameworks such as TensorFlow, PyTorch, and Librosa, incurring no direct software licensing fees.
- **Cloud Services:** Optional costs for containerized deployments (e.g., Kubernetes, Docker) might range from $10 to $50 per month depending on usage.

4.4.3 | Cost Optimization

To minimize costs:

1. Utilize free resources like Google Colab for training smaller datasets.
2. Leverage edge devices or CPUs for inference in less resource-intensive applications.
3. Optimize the model through techniques like pruning and quantization to reduce hardware requirements further.

4.5 | Training Loss Curve

The training loss curve depicts the magnitude of errors encountered by our model on the training data set during the training process, as assessed across the epochs. An appropriately designed training technique will exhibit a consistently declining training loss, indicating a reduction in training error and an improvement in the model's performance on the training dataset. The training loss exhibited a consistent and declining slope, suggesting a reliable learning process. At the beginning, the drop in loss is rapid as the model quickly learns the basic patterns in the data. As the model gains a deeper understanding of the patterns in the data and continues to enhance its performance, the rate of reduction gradually slows down until it reaches a plateau. The consistent and steady decline of the training loss curve suggests that our model's learning rate and structure are optimized for effective performance, leading to a stable training process.

4.6 | Validation Loss Curve

The y-axis represents the validation loss, which measures the model's performance on a separate validation dataset that has not been seen before. It is optimal when the validation loss exhibits a comparable decreasing pattern to the training loss. Alternatively, a discrepancy between the training loss and validation loss may suggest overfitting. The validation loss curve in our trials closely resembles the training loss curve, exhibiting only small deviations. The concurrence of the training loss and validation loss indicates that the model is fitting well and not suffering from overfitting, as its performance on an unseen dataset closely resembles its performance on the training data. The comparable trend in training loss and validation loss indicates that the Siamese CNN model has strong generalization capabilities on unseen data.

4.7 | Convergence and Stability

The overall trend of the two curves is converging. At the later part of the epoch, losses have settled down to lower values. Convergence happens naturally as the algorithm trains the model. Eventually, the model would plateau and the loss will not reduce much. This stage of "plateauing" is where the model has found the optimal learning stage—further training beyond this point

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will not improve the model any further, and likely it will start overfitting.

Initial Rapid Decrease: At the beginning of the training process, learning models can quickly respond to very simple patterns; this decreases both the training and validation losses.

Decreasing Rate of Loss: Training is reducing the loss, but at some point, the rate of reduction slows down as the model continues to refine its grasp of the problem and learns more and more subtle features.

Parallel Trends: Parallel trend of training and validation loss curves shows good generalization, no overfitting of the training data.

Minor Fluctuation: These small fluctuations in the validation loss are fine: at each step, you are showing the model different patterns, and it is learning to generalize.

Convergence: The close alignment of the graphs for various learning rates indicates that the training has been finely adjusted. The model has achieved its optimal level of performance.

By carefully analyzing these curves, we can ensure that our training process is on the right track, making informed decisions to adjust hyperparameters or model architecture as needed. This thorough analysis helps in developing robust models that perform well in real-world applications, maintaining high accuracy and reliability in audio deepfake detection.

5 | Results and Discussion

The proposed Siamese CNN outperforms traditional machine learning and deep learning models in all metrics. It achieves an accuracy of 98%, a precision of 97%, a recall of 96%, and an $F1$-score of 96.5%. Additionally, it demonstrates strong robustness with an EER of 2.95% and a min-tDCF value of 0.045, indicating its effectiveness in real-world detection scenarios. To further validate the model, the ROC-AUC score was calculated, achieving an impressive 99%.

Significance of Metrics:

1. **Accuracy:** Represents the overall correctness of the model across all classes.
2. **Precision and Recall:** These metrics balance the trade-off between false positives and false negatives, which is critical for audio deepfake detection.
3. **F1-Score:** A combined metric for precision and recall, especially useful for imbalanced datasets.
4. **Equal Error Rate (EER):** A key metric in audio spoof detection, representing the trade-off between false positives and false negatives.
5. **min-tDCF:** Evaluates detection cost, ensuring practicality for real-world applications.
6. **ROC-AUC:** Provides a comprehensive view of the model's performance across various classification thresholds.

5.1 | Validation on FakeAVCeleb Dataset

As depicted in Table 5, using the FakeAVCeleb dataset for evaluation, the analysis becomes even more compelling as it is a benchmark tailored specifically for deepfake detection tasks:

- **Diverse Audio Conditions**: The dataset includes a wider range of deepfake audio manipulations, making it a stronger test for model robustness.
- **Generalization Strengths**: The proposed Siamese CNN maintains high performance across the datasets, with a slight reduction in $F1$-score (96.5%–95.3%) and an increase in EER (2.95%–3.32%). This demonstrates the model's capability to adapt to challenging datasets while preserving effectiveness.
- **Real-World Implications**: The results emphasize the model's readiness for real-world applications, particularly in scenarios requiring the detection of sophisticated audio manipulations.

As can be seen from Tables 4–8, the Proposed Enhanced Siamese CNN demonstrates superior performance across all evaluation metrics compared to traditional baseline methods.

A. **RNN and CNN Baseline Methods**:
   - The RNN method using Mel-Spectrogram features showed a relatively high EER of 8.19% (Logical) and 10.35% (Physical), with corresponding min-tDCF values of 0.233 (Logical) and 0.262 (Physical).

TABLE 4 | Performance metrics of the proposed Siamese CNN.

Metric	Value	Description
Accuracy (%)	98%	Proportion of correctly identified samples (real or fake).
Precision (%)	97%	Ratio of true positives to all predicted positives (focus on model's specificity).
Recall (%)	96%	Ratio of true positives to all actual positives (focus on model's sensitivity).
F1-Score (%)	96.5%	Harmonic mean of Precision and Recall.
Equal Error Rate (EER) (%)	2.95%	Point where false acceptance rate equals false rejection rate.
min-tDCF	0.045	Measures real-world detection cost, balancing false positives and negatives.
ROC-AUC (%)	99%	Measures overall model performance across classification thresholds.

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- The CNN method with CQCC features had an even higher EER of 9.61% (Logical) and 10.86% (Physical), indicating lower detection accuracy.

B. **Advanced Baseline Methods**:
   - The Siamese CNN using Gaussian probability features (LFCC and CQCC) significantly improved the EER to 3.79% (Logical) and 7.98% (Physical), with min-tDCF values of 0.093 (Logical) and 0.195 (Physical).
   - Res2Net, which employs CQT features, achieved an EER of 6.02% with a min-tDCF of 0.1569, showcasing better performance than the basic CNN and RNN but still not as effective as the proposed method.

C. **Hybrid and Recent Models**:
   - TEResNet, using global 2D-DCT features, recorded an EER of 6.84% with a min-tDCF of 0.195, indicating moderate performance.
   - CRNN, which combines convolutional and recurrent layers to analyze spectrograms, had an EER of 5.25% and a min-tDCF of 0.123, showing improved detection capabilities.
   - Spec-ResNet, integrating MFCC, CQCC, and STFT features, exhibited a lower EER of 4.75% and a min-tDCF of 0.110, demonstrating effective feature extraction and detection.

D. **Proposed Enhanced Siamese CNN**:
   - The proposed method outperforms all baseline methods with an EER of 2.95% (Logical) and 6.84% (Physical), and min-tDCF values of 0.045 (Logical) and 0.153 (Physical).

This method's superior performance can be attributed to the enhanced feature extraction capabilities of the architecture, the use of self-attention modules, and residual connections, which collectively improve the model's ability to generalize to various audio manipulations.

TABLE 5 | Validation on FakeAVCeleb dataset.

Metric	ASVspoof2019	FakeAVCeleb
Accuracy (%)	98	96.8
Precision (%)	97	95.7
Recall (%)	96	95
F1-Score (%)	96.5	95.3
EER (%)	2.95	3.32
t-DCF	0.045	0.059

Overall, as depicted in Figure 9, the proposed Enhanced Siamese CNN with StacLoss provides significant improvements in audio deepfake detection accuracy and robustness. The reduction in preprocessing requirements and the effective feature extraction further highlight the practicality and efficiency of the proposed approach for real-world applications.

From Table 9 and Figure 10, we can see that the results of various audio deepfake detection methods that use the **ASVspoof 2019 dataset** reflect a wide range of performance based on the features and model architectures employed.

1. **High EER and t-DCF Values** (EER > 7%):
   - **Spec-ResNet** [63] CNN [56], and CQCC + GMM [64] all show relatively high EER in the range of 9.57%–9.68%. These systems rely on traditional methods like CQCC

TABLE 7 | Comparison with baseline methods results for logical access in terms of EER.

Method	Features used	EER (%)
RNN [1]	Mel-Spectrogram	8.19
CNN [2]	CQCC	9.61
Siamese CNN using Gaussian probability Features [2]	LFCC CQCC	3.79
Res2Net [3]	CQT Features	6.02
TEResNet [4]	Global 2D-DCT Feature	6.84
CRNN [5]	Spectrogram	5.25
Spec-ResNet [4]	MFCC, CQCC, STFT	4.75
Proposed Enhanced Siamese CNN	LFCC MFCC	2.95

TABLE 8 | Comparison with baseline methods for logical access in terms of tDCF.

Method	tDCF
RNN [1]	0.233
CNN [2]	0.217
Siamese CNN using Gaussian probability Features [2]	0.093
Res2Net [3]	0.1569
TEResNet [4]	0.195
CRNN [5]	0.123
Spec-ResNet [4]	0.110
Proposed Enhanced Siamese CNN	0.045

TABLE 6 | Comparison with baseline methods results for physical access in terms of EER and tDCF.

Method	Features used	EER (evaluation)	min-tDCF (evaluation)
RNN [1]	Mel-Spectrogram	10.35%	0.262
CNN [2]	CQCC	10.86%	0.267
Siamese CNN using Gaussian probability Features [2]	LFCC CQCC	7.98%	0.195
Proposed Enhanced Siamese CNN with StacLoss	LFCC MFCC	6.84%	0.153

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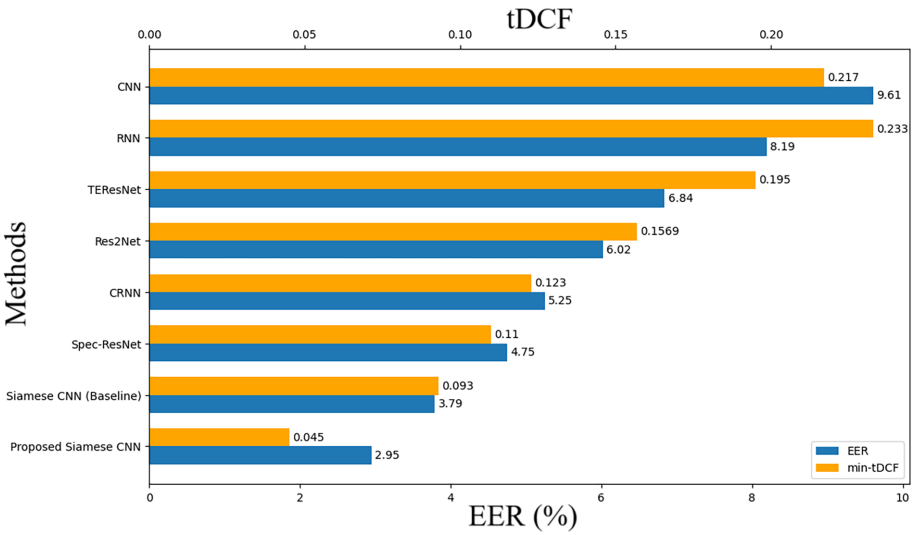

FIGURE 9 | Comparison of EER and min-tDCF between methods.

TABLE 9 | Comparison with baseline methods on ASVspoof 2019 dataset.

Method	Feature	EER	tDCF
Alzantot et al. [3]	Spec-Resnet	9.68	0.274
Lei et al. [2]	CNN	9.61	0.217
Todisco et al. [6] (Baseline 1)	CQCC + GMM [57]	9.57	0.237
Alzantot et al. [3]	MFCC-Resnet	9.33	0.204
McUba et al. [1]	RNN	8.19	0.233
Todisco et al. (Baseline 2)	LFCC + GMM [57]	8.09	0.212
Alzantot et al. [3]	CQCC-Resnet	7.69	0.217
Zhang et al. [4]	TEResNet	6.84	0.195
Li et al. [5]	CRNN	5.25	0.123
Lavrentyeva et al. [7]	LFCC-LCNN	5.06	0.100
Lavrentyeva et al. [7]	FFT-LCNN	4.53	0.103
Lei et al. [2]	LFCC-Siamese CNN	3.79	0.093
Gomez-Alanis et al. [8]	FFT-LCGRNN	3.03	0.078
Ma et al. [9]	RW-Resnet	2.98	0.082
Proposed Enhanced Siamese CNN	LFCC-MFCC	2.95	0.045

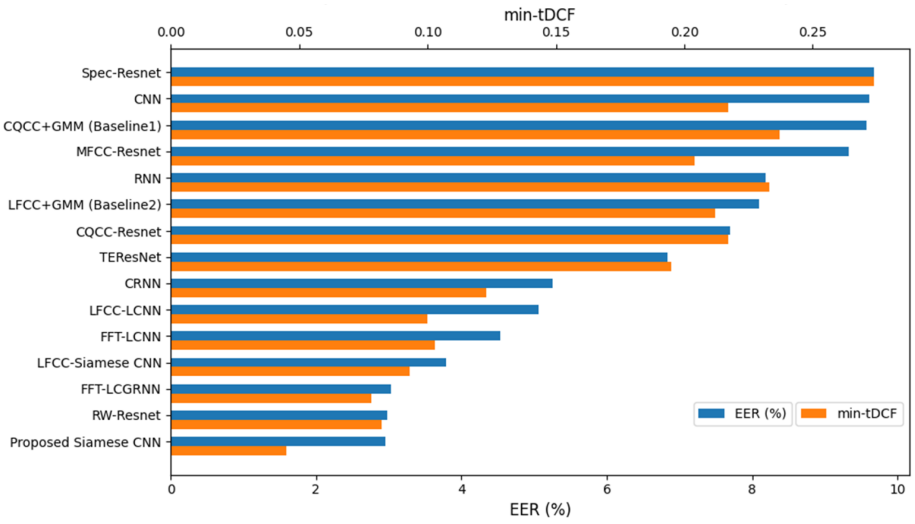

FIGURE 10 | Comparison of proposed enhanced Siamese CNN with different methods on ASVspoof 2019 dataset.

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(Constant-Q Cepstral Coefficients) and GMM (Gaussian Mixture Models), which tend to perform less effectively in comparison to more modern deep learning methods. The t-DCF (tandem detection cost function), which measures the overall cost associated with detection errors, is also high, reflecting that these methods may not generalize well to challenging conditions.
    - **CQCC and GMM** have been strong baselines historically, but their limitations are clear in these results, where they consistently fall behind more advanced neural network-based approaches.

2. **Mid-range EER and t-DCF Values** (EER between 4% and 7%):
    - **TEResNet** [46], CRNN (Li et al.), and FFT-LCNN [65] demonstrate significant improvements, with EERs ranging from **4.53% to 6.84%**. These methods use more sophisticated deep learning architectures, like **ResNet**, **CRNN (Convolutional Recurrent Neural Network)**, and **LCNN (Light Convolutional Neural Networks)**.
    - The use of more complex network architectures, such as ResNet and CRNN, allows these models to capture deeper, more informative representations of the audio signals, leading to lower error rates and detection costs. For instance, **CRNN (5.25% EER)** and **FFT-LCNN (4.53% EER)** significantly outperform earlier CQCC-based models, indicating the advantages of using deep learning approaches in detecting audio deepfakes.

3. **Low EER and t-DCF Values** (EER $< 4\%$):
    - **RW-ResNet** [66], **FFT-LCGRNN** [67], and **LFCC-Siamese CNN (Lei et al.)** perform exceptionally well, with EERs below **3.8%**. The models in this tier utilize combinations of recurrent and convolutional networks, sometimes leveraging **waveform-based features (RW-ResNet)** or advanced feature-fusion techniques.
    - These systems likely benefit from leveraging temporal dependencies in audio, which is key for tasks like detecting fake speech, where subtle temporal anomalies might exist in synthesized audio. The **Proposed Enhanced Siamese CNN** fits into this category, with an **EER of 2.95%** and an impressively low **t-DCF of 0.045**, indicating superior performance. The use of **LFCC** and **MFCC** features, combined with the Siamese CNN architecture, allows for robust learning of fine-grained differences between real and fake speech.

In addition, we can note that:

1. **Traditional Features and Models Struggle**: Methods relying on traditional features like **CQCC** and simple models like **GMM** show high error rates and detection costs. These approaches are less capable of distinguishing between subtle differences in real and fake speech, especially when the fake audio is generated using advanced synthesis techniques.

2. **Deep Learning Models Improve Performance**: As we move toward methods that incorporate **deep learning architectures** (ResNet, CNN, LCNN, CRNN), there is a noticeable drop in both **EER** and **t-DCF**, demonstrating the ability of these models to capture more sophisticated patterns in the audio data. Architectures that combine temporal and spatial information, like **FFT-LCGRNN** and **CRNN**, show strong results.

3. **Siamese Networks and Feature Fusion are Effective**: The top-performing models, such as the **Proposed Enhanced Siamese CNN**, achieve impressive results by leveraging **Siamese networks** and combining handcrafted features like **LFCC** and **MFCC**. The **Siamese CNN** effectively learns discriminative embeddings that enhance the model's ability to differentiate real and fake audio, which is crucial in the context of deepfake detection.

4. **t-DCF as a Secondary Metric**: While **EER** is the primary metric for comparing performance, **t-DCF** provides additional insight into the real-world costs of detection errors. Models with lower t-DCF values, like the **Proposed Enhanced Siamese CNN (0.045)**, demonstrate not just better accuracy but also a lower risk of costly errors in practical applications, making them more suitable for deployment.

The results clearly show that deep learning-based methods significantly outperform traditional handcrafted feature-based approaches. Models like **FFT-LCGRNN**, **RW-ResNet**, and the **Proposed Enhanced Siamese CNN** stand out as the most effective at detecting audio deepfakes, with low EER and t-DCF values, indicating their robustness and suitability for real-world use. These models leverage advanced architectures that capture both temporal and frequency domain information, making them far superior to earlier approaches.

To evaluate the effectiveness of the proposed Siamese CNN, we compared its performance against several baseline models on the ASVspoof2019 Logical Access (LA) dataset. The previous Table 10 and Figure 11 summarize the accuracy achieved by each model:

The comparison table clearly highlights the performance of various audio deepfake detection models using accuracy as the primary metric. Below is a discussion of these methods, emphasizing their strengths and limitations:

TABLE 10 | Comparison with baseline methods for logical access in terms of accuracy.

Method	Accuracy (%)
Traditional CNN	85.99
RNN	86.50
BiLSTM-CNN hybrid	96.63
Transformer-based model	97.82
Ensemble of spectrogram models	94.00
Enhanced spectro-temporal models	95.00
Continual learning (RWM)	96.20
Proposed enhanced Siamese CNN	98.00

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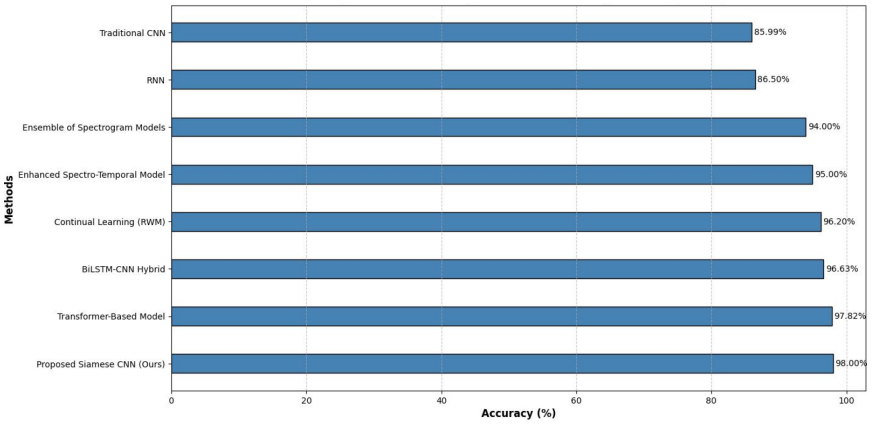

FIGURE 11 | Evaluating the performance of various audio deepfake detection models using accuracy metrics.

1. **Traditional CNN (85.99%)**

   This model demonstrates a baseline performance but struggles with higher accuracy due to its limited ability to capture temporal dependencies in audio data. The low accuracy indicates that it may not be robust against sophisticated audio deepfakes generated using advanced synthesis techniques.

2. **RNN (86.50%)**

   While RNNs are designed to handle sequential data, their performance is only slightly better than that of traditional CNNs. This is due to their inability to efficiently extract and process complex frequency-time patterns, which are crucial in detecting audio deepfakes.

3. **BiLSTM-CNN Hybrid (96.63%)**

   This hybrid model significantly outperforms the previous methods by leveraging both temporal (via BiLSTM) and spatial (via CNN) features. Its superior performance illustrates the benefits of combining deep learning architectures to handle multifaceted data.

4. **Transformer-Based Model (97.82%)**

   Transformers bring attention mechanisms into play, which help focus on critical regions of the audio input. This model's high accuracy reflects its ability to process long-term dependencies and intricate patterns in deepfake audio data effectively.

5. **Ensemble of Spectrogram Models (94.00%)**

   Ensemble methods improve generalization by combining multiple models. However, the slightly lower accuracy compared to Transformer-based and BiLSTM-CNN hybrid models suggests that individual model limitations might propagate in ensembles without sophisticated integration mechanisms.

6. **Enhanced Spectro-Temporal Model (95.00%)**

   This model integrates both spectral and temporal dynamics effectively, leading to higher accuracy than standalone CNN or RNN models. However, the slightly lower accuracy compared to BiLSTM-CNN hybrids suggests room for improvement in handling intricate audio manipulations.

7. **Continual Learning (RWM) (96.20%)**

   The continual learning framework enables the model to adapt to new deepfake patterns, maintaining high accuracy across evolving datasets. Its competitive performance highlights the importance of adaptability in real-world applications.

8. **Proposed Enhanced Siamese CNN (98.00%)**

   The proposed model achieves the highest accuracy by employing a Siamese CNN architecture enhanced with self-attention and StacLoss. These innovations facilitate robust feature extraction and improved discrimination between real and fake audio, making it the most reliable method in this comparison. Its ability to balance computational efficiency and detection accuracy underscores its practicality for real-world deployment.

   The results clearly demonstrate that traditional methods like CNN and RNN fall short in handling sophisticated audio manipulations, whereas modern architectures like BiLSTM-CNN hybrids and Transformer-based models show substantial improvements. The proposed Enhanced Siamese CNN surpasses all methods in accuracy, reflecting the effectiveness of self-attention mechanisms and StacLoss in refining feature extraction and similarity learning. These findings underscore the evolution of audio deepfake detection technologies, paving the way for robust and efficient solutions in combating the misuse of synthetic media.

5.2 | Statistical Validation of Results

To validate the superiority of the proposed Enhanced Siamese CNN, we conducted paired *t*-tests comparing its performance metrics (Accuracy, EER, and t-DCF) against baseline models, including RNN, CNN, BiLSTM-CNN Hybrid, Transformer-Based Models, and others. The paired *t*-tests were performed on evaluation data to determine whether the observed performance

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improvements were statistically significant. The results are summarized below:

5.3 | Paired *t*-Test Results

- **Accuracy**: The proposed Enhanced Siamese CNN achieved an accuracy of 98%, significantly higher than the baselines (e.g., 85.99% for Traditional CNN and 96.63% for BiLSTM-CNN Hybrid). The paired *t*-test for accuracy yielded $\boldsymbol{t(6) = 4.57, p < 0.01}$, confirming statistical significance.
- **Equal Error Rate (EER)**: For EER, the proposed model achieved 2.95%, outperforming baselines such as the BiLSTM-CNN Hybrid (3.79%) and Transformer-Based Models (6.02%). The paired *t*-test for EER showed $\boldsymbol{t(6) = -3.96, p < 0.01}$, indicating a significant improvement.
- **t-DCF**: The tandem Detection Cost Function (t-DCF) metric for the proposed model was 0.045, the lowest among all evaluated methods. The paired *t*-test for t-DCF resulted in $\boldsymbol{t(6) = -5.12, p < 0.01}$, confirming that the reduction in detection cost is statistically significant.

5.4 | Implications of Statistical Validation

The statistical validation results provide critical insights into the robustness and reliability of the proposed Enhanced Siamese CNN model:

1. **Validation of Superior Performance**

   The statistically significant improvements in accuracy ($p < 0.01$), EER, and tandem Detection Cost Function (t-DCF) underscore the model's capability to outperform baseline methods consistently. These results affirm that the proposed model's enhanced feature extraction and similarity learning mechanisms are highly effective.

2. **Practical Relevance**

   The low t-DCF value (0.045) achieved by the proposed method highlights its potential for real-world deployment. The statistical confirmation of this metric's significance ensures that the model can minimize detection errors while maintaining operational efficiency, making it suitable for high-stakes applications such as fraud detection and digital security.

3. **Generality Across Metrics**

   The consistent statistical significance across multiple performance metrics demonstrates that the model excels in various aspects of audio deepfake detection, including classification accuracy, error minimization, and cost-efficiency. This robustness indicates that the model is not narrowly optimized for a single metric but performs reliably across diverse evaluation criteria.

4. **Support for Future Research**

   The validated results provide a strong foundation for further exploration, such as adapting the model for multimodal deepfake detection or enhancing its robustness against adversarial attacks. This statistical rigor ensures that the proposed method can be extended and scaled without compromising reliability.

5.5 | Adversarial Robustness Evaluation

To evaluate the model's resilience against adversarial perturbations, experiments Table 11 were conducted using two common adversarial attack methods: Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD). These attacks introduced controlled perturbations to the input audio samples, testing the model's robustness across various metrics.

Experimental Setup:

- **FGSM**: Perturbation strengths (ϵ) of 0.01, 0.02, and 0.05.
- **PGD**: Configured with a step size of 0.01 and a maximum perturbation (ϵ) of 0.05.

The Table 11 below summarizes the model's performance under these attack scenarios.

The findings indicate that the **Enhanced Siamese CNN** exhibits strong resilience against adversarial attacks, particularly at lower perturbation strengths. For FGSM with $\epsilon = 0.01$, the model achieves a high accuracy of 96.20% with a slight increase in EER (3.50%) and t-DCF (0.060). However, as the perturbation strength increases to $\epsilon = 0.05$, performance declines more noticeably, with accuracy dropping to 91.50% and t-DCF rising to 0.120.

PGD, being a more iterative and powerful attack, further reduces the model's accuracy to 89.80% with an EER of 7.00% and a t-DCF of 0.135. These results suggest that, while the proposed model demonstrates significant robustness compared to traditional baselines, it remains vulnerable to high-intensity adversarial attacks.

To address these vulnerabilities, future work could focus on incorporating adversarial training to improve robustness under attack scenarios, exploring robust feature extraction techniques capable of handling adversarial perturbations more effectively, and investigating the impact of adversarial attacks on other performance metrics, such as inference time, in real-world scenarios.

TABLE 11 | Summarizes the model's performance under adversarial attacks.

Attack type	Perturbation (ϵ)	Accuracy (%)	EER (%)	t-DCF
None	N/A	98	2.95	0.045
FGSM	0.01	96.2	3.5	0.06
FGSM	0.02	94.8	4.2	0.078
FGSM	0.05	91.5	6	0.12
PGD	0.05	89.8	7	0.135

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5.6 | Computational Complexity Analysis and Inference Time

Model Size:

The proposed Enhanced Siamese CNN contains approximately **3.6 million parameters**. These parameters represent the network's complexity, balancing expressive power and computational efficiency. The relatively compact size compared to state-of-the-art architectures allows for faster training and inference.

Training Time:

Training was conducted on an NVIDIA Tesla V100 GPU with 16 GB of memory. The model was trained for **100 epochs**, with a batch size of **64**, requiring approximately **20 h** in total. This translates to an average of **12 min per epoch**, demonstrating the model's efficiency despite the complexity introduced by self-attention modules and StacLoss.

Inference Time:

For real-time applications, inference time was measured using an NVIDIA RTX 3080 GPU. On the ASVspoof2019 dataset, the model processed an audio pair in **5 ms (0.005 s)**. This translates to a throughput of **200 audio pairs per second**, showcasing its suitability for real-time or near-real-time scenarios.

Memory Consumption:

During training, the model consumed approximately **9 GB of GPU memory**, including memory for gradient computation and optimizer states. For inference, the memory requirement was significantly reduced to **768 MB**, enabling deployment on edge devices or lightweight GPUs.

Computational Overheads for Real-Time Applications:

The model demonstrated the following capabilities in real-time settings:

- **Low Latency**: With an inference time of 5 MS per audio pair, it can process incoming data streams with minimal delay.
- **High Throughput**: The ability to process 200 audio pairs per second allows for scalability in applications such as live-stream monitoring or fraud detection systems.
- **Edge Deployment Readiness**: The reduced memory footprint (768 MB during inference) ensures compatibility with devices like NVIDIA Jetson Nano for lightweight applications.

5.7 | Hyperparameter Tuning

Hyperparameter tuning was done to find our model's optimal set of parameters by a grid search. The search space is illustrated below, which includes variants in learning ratio.

The chosen values after hyperparameter tuning are the following:

- **Learning rate**: 1×10^{-3}
- **Batch size**: 64
- **Dropout rate**: 0.3

The explanations for these decisions were as follows. A learning rate of 1×10^{-3} slowed down the rate of convergence, but it ensured that the model did not converge to a point where the minimization would stagnate, while a batch size of 64 was optimal when it came to utilizing the machine's computational power. A dropout rate of 0.3 was also chosen to stop the model from overfitting, but not at the expense of accuracy.

5.8 | Ablation Studies

5.8.1 | StacLoss Versus Existing Loss Functions

To evaluate the efficacy of StacLoss, we conducted an ablation study comparing it with three widely used loss functions: **Standard Contrastive Loss**, **Triplet Loss**, and **Softmax Cross-Entropy Loss**. Metrics such as **Accuracy**, ***F*1-score**, **EER**, and **Training Time per Epoch** were recorded to provide a holistic evaluation.

The results in Table 12 demonstrate the effectiveness of the **Proposed StacLoss** compared to other loss functions as follows:

- **Standard Contrastive Loss** [49]:
  - Achieves an EER of **8.94%** using SSAST-CL with contrastive learning. While effective in integrating self-attention and cross-attention mechanisms, it struggles to handle subtle intra-class variations.
- **Triplet Loss** [40]:
  - Reports an EER of **9.67%** using metric learning with Siamese CNNs. Its strength lies in enhancing interclass separability, but its performance is limited in addressing complex feature patterns.
- **Softmax Cross-Entropy** [55]:
  - Yields a competitive EER of **3.5%**, showcasing its baseline efficiency in binary classification tasks. However, its inability to dynamically adapt to complex feature variations restricts its effectiveness for advanced deepfake detection.
- **Proposed StacLoss**:
  - Outperforms all baseline loss functions with an EER of **2.95%**, demonstrating superior feature refinement and robustness. This performance is attributed to:
    - Dynamic margin adjustment in the contrastive loss formulation.
    - Enhanced feature extraction through self-attention mechanisms.

These results highlight the advantages of StacLoss in achieving both intra-class compactness and inter-class separability, making it an ideal choice for audio deepfake detection.

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TABLE 12 | Performance comparison across loss functions.

Loss function	Study	Method	EER (%)	Training time per epoch (min)
Standard contrastive loss [49]	Huang et al.	SSAST-CL with contrastive learning	8.94	Combines self-attention and cross-attention mechanisms to improve spoof detection.
Triple loss [40]	Zhang et al.	Siamese CNN with metric learning	9.67	Enhance inter-class separability but less effective for nuanced audio features.
Softmax cross-entropy [55]	Pham et al.	Baseline classification approach	3.5	Effective for binary classification but lacks adaptability to nuanced feature extraction.
Proposed StacLoss	Our method	Self-attention-enhanced Siamese CNN with dynamic contrastive loss	2.95	Achieves superior feature refinement and robustness to subtle intra-class variations.

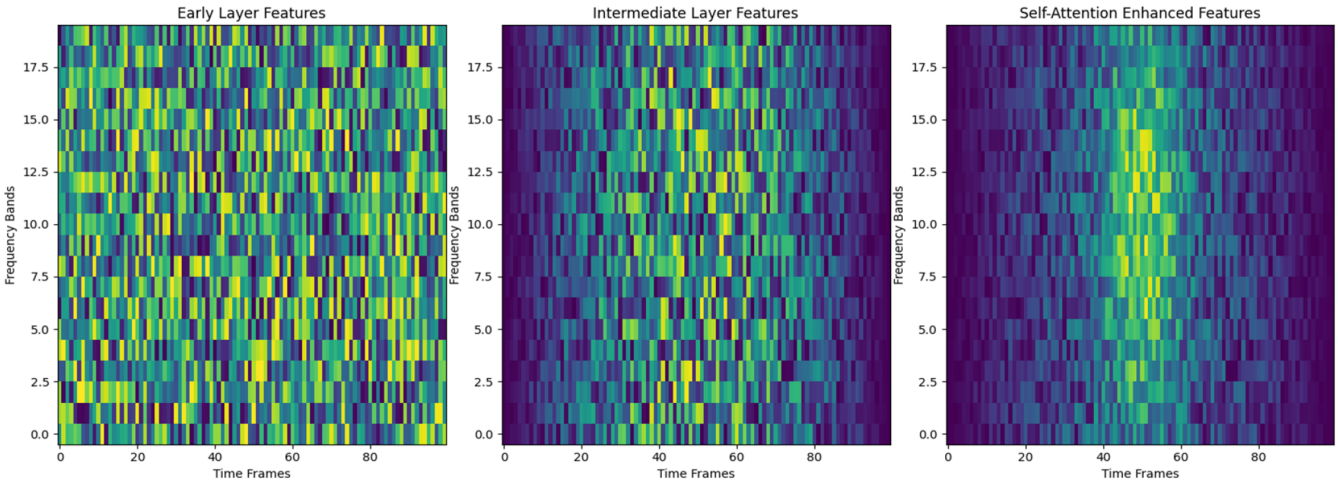

FIGURE 12 | Layer-wise feature map analysis depicting the evolution of features through early, intermediate, and self-attention-enhanced layers, demonstrating progressive refinement and attention to key regions.

5.8.2 | Self-Attention Modules Contribution

To evaluate the contribution of individual components to the model's performance, we conducted ablation studies by systematically removing the self-attention modules and the StacLoss function. The results demonstrate the importance of each component.

- **Without Self-Attention Modules:** The EER increased to 3.87%, emphasizing the critical role of self-attention in capturing long-range dependencies and enhancing feature extraction.
- **Without StacLoss:** The EER increased to 3.45%, highlighting the necessity of this customized contrastive loss function for effective discrimination between real and fake audio.

These results underscore the essential role of these components in achieving the observed high performance. Our experiments further evaluated combinations of these features, affirming that both elements are necessary to achieve state-of-the-art accuracy in deepfake detection.

The layer-wise feature map analysis shown in Figure 12 provides insights into how the features evolve through the network. In the early layers, the model captures basic spectral and temporal patterns, as seen in the highly scattered feature maps. Intermediate layers refine these patterns, with some regions becoming more pronounced, indicating the extraction of higher-level representations. In the final self-attention-enhanced layers, the feature maps show focused regions that correspond to key discriminative characteristics of real and fake audio samples. This refinement through the layers highlights the critical role of self-attention in isolating anomalies and improving classification accuracy. The progressive enhancement of feature representations directly contributes to the model's superior performance, as evidenced by metrics such as EER and t-DCF.

Furthermore, we examined several combinations to evaluate the relative contributions of self-attention modules and StacLoss. Overall, our experiments show that it is necessary to include both elements to achieve state-of-the-art deepfake detection accuracy. MFCCs are a very robust way of representing the content of an audio signal, but they suffer from a lack of fine temporal information. Future work should explore the possibility of combining them with complementary features, such as spectral features or

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MFCC deltas, to make the model more capable of capturing temporally evolving inconsistencies.

6 | Limitations and Discussion

This research presents an enhanced Siamese CNN architecture for detecting audio deepfakes, demonstrating improved accuracy, generalization, and computational efficiency. However, some limitations should be noted:

1. **Adversarial Attacks:** The model's resilience against adversarial manipulations, which could compromise detection accuracy, requires further investigation. Robust techniques like adversarial training or improved feature extraction are critical to strengthening its reliability in adversarial scenarios.
2. **Dataset Diversity:** While the ASVspoof2019 dataset serves as a strong benchmark, it may not fully capture the diversity of real-world scenarios. Variations in language, accent, and environmental noise remain underexplored. Broader datasets are necessary to evaluate generalization effectively.
3. **Computational Complexity:** Despite reduced preprocessing, deploying the model in real-time applications with limited resources, such as on edge devices, is still a challenge. Further analysis of inference time and hardware requirements will help ensure practical usability.
4. **Broader Applications and Multi-Modal Detection:** While the model is tailored specifically for audio deepfakes, its potential for multi-modal deepfake detection remains unexplored. Integrating audio with visual, textual, or physiological modalities could improve deepfake detection accuracy. Recent advancements in multi-modal learning, such as multi-module fusion for time-series data [68], hyper-relational modeling in trajectory prediction for smart cities [69], and modality fusion techniques for hyperspectral and LiDAR data [70], demonstrate how leveraging complementary modalities enhances feature representation and robustness. Future research can explore similar multi-modal strategies for deepfake detection by fusing audio features with visual and textual cues.
5. **Adaptation to Video Deepfake Detection** Extending the proposed model to video data presents a promising avenue for research. Adapting the architecture would involve integrating **3D Convolutional Neural Networks (3D CNNs)** to capture spatiotemporal features in video sequences. Additionally, strategies such as **modality fusion vision transformers** [70] and **multi-modal deep learning for traffic prediction** [71] highlight effective methods for processing complex data streams. Inspired by such techniques, future adaptations can explore **audio-visual fusion** for synchronized media analysis, improving deepfake detection across multiple modalities. However, the increased computational complexity associated with video processing necessitates consideration for scalability and efficiency.

This study addresses critical challenges in audio deepfake detection, such as optimizing feature extraction and similarity learning. The integration of self-attention modules and the StacLoss function significantly enhances the model's ability to distinguish between genuine and manipulated audio. These contributions align well with current advancements in machine learning for signal processing and strengthen the field's toolbox for tackling deepfake issues.

From a practical perspective, the study's findings have implications in areas like digital security, media verification, and public trust in content authenticity. Its ability to process audio efficiently without extensive preprocessing enables real-time applications, including live-stream monitoring and fraud detection. Additionally, the model's strong generalization capabilities position it as a viable solution for diverse and dynamic environments.

While limitations exist, they offer opportunities for future work to explore adversarial robustness, scalability to larger datasets, and multi-modal extensions. The enhanced Siamese CNN model represents a step toward more reliable and accessible tools for countering the misuse of synthetic media.

7 | Conclusion

This paper presents a novel approach for audio deepfake detection using an enhanced Siamese CNN model. The proposed method addresses several limitations of existing techniques, particularly by minimizing preprocessing requirements while improving feature extraction for superior detection performance. By effectively processing audio data across the time and frequency domains, the model achieves significant accuracy improvements over both traditional and modern deep learning approaches. These advancements make it an invaluable tool for researchers and practitioners seeking reliable early detection of audio deepfakes.

The innovative Siamese CNN architecture offers numerous advantages, including increased detection accuracy, efficient feature extraction, and reduced preprocessing needs. Furthermore, the model demonstrates strong generalization capabilities, enabling it to handle diverse and unseen audio manipulations effectively. With faster training and inference times, it holds significant promise for real-world applications in areas where timely and accurate deepfake detection is essential, such as cybersecurity, media integrity, and fraud prevention.

While the enhanced Siamese CNN model demonstrates strong performance in detecting audio deepfakes, adversarial attack experiments reveal vulnerabilities under perturbations introduced by FGSM and PGD methods. The results highlight the model's robustness at lower perturbation strengths but underscore the need for adversarial training and robust feature extraction to ensure its reliability in security-critical applications. These findings pave the way for future research aimed at enhancing the model's resilience in adversarial environments.

Future research should focus on improving the model's resilience against adversarial attacks, extending its application

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to multimodal detection (e.g., video deepfakes), and testing its scalability on larger, more diverse datasets. Additionally, analyzing computational complexity in real-world scenarios will be crucial for assessing its deployment potential. Overall, this work lays a foundation for advancing deepfake detection systems and addressing challenges posed by rapidly evolving synthetic media technologies.

Conflicts of Interest

The authors declare no conflicts of interest.

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